

Collins ep reverse-unitarity analytical comparison: v0.5.1

ANALYTICAL_REVIEW_COMPLETE_WITH_QUALIFICATIONS. Actual exports from two frozen-source official r00-r07 workflows are available. This equation review reads final run A and preserves historical statuses. Seeded verification, dependency checks and pair acceptance are reported separately in the supporting numerical report. Incomplete whole-group comparisons and the independently checked literal K11 bulk discrepancy remain explicit. **Reading this report.** New native expressions, accepted-engine expressions and literal paper equations occupy separate columns. Historical status is preserved without revision. Current exact differences are recomputed from the supplied native records; empty rows remain unverified. The 167 validator entries are scalar comparisons, not 167 literature equations. All numerical checks are confined to the supporting report.

Equation-group scope	EXACT	AFTER EXPLICIT CONVERSION	MISMATCH	UNVERIFIED
Historical groups, unchanged	14	35	4	32
New versus accepted engine	42	8	0	35
New versus published target	0	23	0	62

Separately checked C16 subexpressions: actual final HF bulk agrees with converted YZ and has a nonzero literal K11 C_A residual. That **MISMATCH** is displayed in C16 with an exact admissible test action. Whole-group C16 remains UNVERIFIED for unmatched normalization/auxiliary fields; this does not erase the checked bulk discrepancy. **Assumptions and convention maps.** Massless on-shell external partons. The elastic Born/hard Mandelstam map alone uses $s>0$, $t,u<0$ and $s+t+u=0$; tagged real-radiation photon-parton invariants are defined separately; positive spacelike Q^2 and scales; $D=4-2\epsilon$; $a=\alpha_s/(2\pi)$. Positive-energy particle cuts, measurement deltas and theta boundaries are distinct. Eta expansion and cancellation at fixed epsilon precede epsilon expansion. Physical external transverse spin and the declared common angular/Fourier continuation are explicit conventions. Distribution comparisons retain delta, plus and regular coefficients. **Scope.** Leading-power Born plus one loop, with the inherited narrow-cone and physical HF endpoint qualifications. Generic contact coefficients, $R=1$, unrestricted nonsingular fixed-order NLO, closed full twist-three evolution and NNLO are not certified. Fitted functions remain symbolic and Figure6 remains paused. Shared generating source and retained evidence are disclosed; no analytical equality alone certifies the final physical prediction. **Literal sources.** Pinned source snippets, enclosing blocks, macro definitions, source archives and PDFs are preserved under references/retained. The literal equation column retains the printed symbols and signs. Rendering removes only comments, labels and display alignment. Conversions appear separately. Archived electromagnetic-prefactor, printed Sudakov and literal CA-kernel discrepancies are not silently repaired. Published-target arithmetic is independently preserved from the prior source audit and is rechecked against new readback; an internal identity or context citation is never upgraded to a complete published equation. [Complete convention map](#); [row/source bindings](#); [scientific review](#).

Current derivation evidence and qualifications

The tables distinguish generated native expressions, review-only consequences, accepted expressions and literal references. Native status, exact supporting tests, numerical comparisons and dependency probes are separate evidence categories. An export label is not proof of an independent derivation.

Physical measured family and native sphere. With $p^2 = n^2 = 0$, $p \cdot n = 1$, $\alpha = k \cdot n$, $\beta = k \cdot p$ and $K = k^2$, the implemented family is

$$D_0 = K - 2u\beta, \quad D_1 = \alpha - 1, \quad D_2 = K - 2\beta + \tau, \quad \rho = 2\alpha\beta - K, \quad \rho - \tau = 2\beta D_1 - D_2.$$

Only $D_0 = (k - up)^2$ is a positive-energy emitted-particle cut; the others fix longitudinal fraction and radius. For $0 < u < 1$, $\tau > 0$, the unit-cut Jacobian is $1/[2(1 - u)]$, giving

$$\int d^{4-2\epsilon} k \delta_+(D_0) \delta(D_1) \delta(D_2) = \frac{\pi^{1-\epsilon} \tau^{-\epsilon}}{2(1-u)\Gamma(1-\epsilon)}.$$

Raised measurement deltas require the separately generated derivative-distribution transport; the unit-delta identity alone is insufficient. Actual audited Kira rules have master $M(1, 1, 1)$ and diagnostic raised-target relations

$$M(1, 1, 2) = \frac{4-d}{2\tau} M(1, 1, 1), \quad M(1, 2, 1) = \frac{4-d}{2(u-1)} M(1, 1, 1), \quad M(2, 1, 1) = 0.$$

The physical plane plus -2ϵ evanescent sphere yields a genuine Euler integral,

$$\int_0^\infty dy y^{-1-\epsilon} (1+y)^{\epsilon-1} = B(-\epsilon, 1), \quad N = \frac{e^{\gamma_E \epsilon}}{\Gamma(-\epsilon)} B(-\epsilon, 1) = \frac{e^{\gamma_E \epsilon}}{\Gamma(1-\epsilon)} = 1 - \frac{\pi^2}{12} \epsilon^2 + O(\epsilon^3).$$

Assembly consumes the evaluated native N through its reduction/contraction node. This independently checked Gamma identity is not a replacement producing input. Beam routing uses emitted $l = k - up$ and active $q = p - l$, $q^2 = -\tau/(1 - u)$; its spacelike spin/color traces are generated separately from fragmentation.

Soft connection and fragmenting-jet replacement. At $u = 0$, restoring $a = k \cdot n > 0$ gives $da/a = dy$. For $p = (1, 0, 0, 1)$, $n_J = (\cosh Y, 1, 0, \sinh Y)$ and $r_y = e^{y-Y}$, the generated eikonal current satisfies

$$J^\mu = \frac{p^\mu}{p \cdot k} - \frac{n_J^\mu}{n_J \cdot k}, \quad -J^2 = \frac{4r_y^2}{k_T^2(r_y^2 - 2r_y \cos \phi + 1)}.$$

Conditional angular weights, regulator prescriptions, cone boundaries and overlap allocations are derived separately from the common radial sphere. Eta cancellation is performed at fixed epsilon. Fixed local counterterms preserve the unperturbed unit normalization; nonphysical probes propagate changed masters with the same subtraction recipes.

The extra fragmenting-jet factor is the measured common-operator minus overlap difference, not the inclusive quark jet function. Its zero labels for counterterm, epsilon projection, finite remainder, anomalous-dimension difference and independent finite remainder are consequences/aliases of common-operator equality plus separately checked rapidity integrals. They are not five independent Kira calculations. Nonzero bare and overlap terms remain visible. The inherited narrow-cone qualification excludes unrestricted $R = 1$ certification.

Ordinary virtual current and initial inclusive pilot. The actual ordinary current bubble and finite hard coefficient are

$$B(\epsilon) = e^{\gamma_E \epsilon} \frac{\Gamma(\epsilon) \Gamma(1-\epsilon)^2}{\Gamma(2-2\epsilon)} = \frac{1}{\epsilon} + 2 + \left(4 - \frac{\pi^2}{12}\right) \epsilon + O(\epsilon^2), \quad h_1 = C_F \left(-L_Q^2 - 3L_Q - 8 + \frac{\pi^2}{6}\right).$$

Here $H = 1 + \alpha_s h_1/(2\pi)$ and $L_Q = \ln(\bar{\mu}^2/Q^2)$; vertex and external-leg UV residues remain separate from IR poles. The initial inclusive pilot instead uses

$$N_{\text{pilot}} = e^{\gamma_E \epsilon} \frac{\Gamma(1-\epsilon)}{\Gamma(2-2\epsilon)} = 1 + 2\epsilon + \left(4 - \frac{\pi^2}{4}\right) \epsilon^2 + O(\epsilon^3), \quad C_F \left(-\frac{2}{\epsilon} + \frac{5}{2} - \frac{\epsilon}{2}\right)$$

for its master and integrated angular ratio. That longitudinal moment is distinct from the fixed-fraction measured family. Its success alone establishes no TMD/HF completion.

Spin continuation, physical endpoints and observable. The declared scalar/rank-one continuation divides the full transverse Fourier result by $\Gamma(1 - \epsilon)\Gamma(1 + \epsilon)$. It is not an unconditional replacement of a full-dimensional tensor average by a physical-plane average. The actual transversity angular difference and the separately executed rank-one HF difference are

$$\Delta P_{h_1} = \frac{C_F \epsilon^2 (z - 1)}{1 - \epsilon}, \quad \Delta \rho_{HF} = -\epsilon^2 (1 - u)^2 \frac{C_A - 2C_F + 2C_F v}{(1 - v)^2}.$$

Finite one-loop shifts vanish after the single radial pole on the uniform physical endpoint class. The HF proof integrates the rank-one phase before evanescent-angle averaging. Neither result proves all-epsilon, higher-order or generic-contact equality. Double-zero/regularity assumptions annihilate physical pole/derivative contacts; generic contact coefficients remain undetermined. The zero-virtual HF statement is a whole-branch translation/Wilson-line proof, not an assertion that individual unintegrated loop diagrams vanish.

The conventions explicitly retain

$$p_\perp = -zk_\perp, \quad \hat{H}_3 = -2zM_h H_{1,\text{Trento}}^{\perp(1)}, \quad \tilde{H}^\alpha = -\frac{ib^\alpha}{2z} \hat{H}_3, \quad \frac{b_q db_q}{2\pi} J_0(b_q q_T), \quad \frac{b_h db_h}{2\pi z^2} J_0(b_h j_T/z), \quad \frac{b_h^2 db_h}{4\pi z^3} J_1(b_h j_T/z).$$

Native and Kang raw cuts differ by two in both the diagonal moment and HF function, leaving relative mixing-basis factor one. Printed upper-index, Wilson-link placement and literal C_A -kernel discrepancies remain explicit. No silent operator repair follows from coefficient equality.

The observable uses strict first order in $a = \alpha_s/(2\pi)$ before common flavor/Fourier integration. Lower-limit plus convolutions retain their boundary tails and $d\xi/\xi$ Jacobian. The all- b Collins operator already includes HF; the separate HF action is added only in the local small- b expansion. The physical asymmetry is the ratio after common linear integration. Fitted functions remain symbolic; Figure 6 remains paused. Scale derivatives displayed in the tables are labeled review-only consequences of generated matching and collinear evolution; they are not separately generated anomalous-dimension amplitudes.

Additional executable evidence: [measured wrapper review](#); [inclusive pilot](#); [virtual readback](#); [final measured checks](#); [HF angular proof](#); [final soft review](#). Every table assignment has separate native source/export bindings.

Reverse-unitarity method: literal source and separately stated correction. The pinned method reference is Anastasiou and Melnikov, [hep-ph/0207004v3](#). These displays are included directly from the preserved source snippets; original macros and enclosing source blocks are retained. The Eq. (42) snippet also preserves its trailing source fragment. **Literal Eq. (17):**

$$2i\pi\delta\left(p^2 - m^2\right) \rightarrow \frac{1}{p^2 - m^2 + i0} - \frac{1}{p^2 - m^2 - i0}. \quad (17)$$

Literal Eq. (42):

$$(1 - z)^{-1+a\epsilon} = \frac{1}{a\epsilon} \delta(1 - z) + a\epsilon \left[\frac{1}{1 - x} \right]_+ + \frac{(a\epsilon)^2}{2!} \left[\frac{\log(1 - z)}{1 - z} \right]_+ + \frac{(a\epsilon)^3}{3!} \left[\frac{\log^2(1 - z)}{1 - z} \right]_+ + \mathcal{O}(\epsilon^4), \quad (42)$$

facilitating the cancellation between real and virtual

The following independently checked identities are separate proposed corrections, not substitutions in the literal source above:

$$\delta(x) = \frac{1}{2\pi i} \left[\frac{1}{x - i0} - \frac{1}{x + i0} \right], \quad (1 - z)^{-1+a\epsilon} = \frac{\delta(1 - z)}{a\epsilon} + \sum_{n=0}^{\infty} \frac{(a\epsilon)^n}{n!} \left[\frac{\log^n(1 - z)}{1 - z} \right]_+.$$

The printed discontinuity ordering has the opposite sign to the implemented convention. The first plus coefficient, shifted powers and x versus z in Eq. (42) are unchanged above. Original pages were visually verified in the preserved source audit. Positive cut indices alone prove neither energy support nor absence of differentiated measurement/regulator boundary terms. These method comparisons add no groups to the retained 85-row inventory.

Source archive SHA256: [c42ff3dc5e6941d329a66251c5e292f628866b06c12382b54eebe34761bed0fd](#). PDF SHA256: [f7c43113d55ed6b370450bc86c11667aed18dbe7e14bff52d442611ab8087240](#). Exact snippet and macro hashes are in [references/hep-ph0207004v3/manifest.json](#).

Definitions, kinematics and Born normalization

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
B01 Born UU / α^2	$\frac{2s^2}{t^2} + \frac{2u^2}{t^2}$	$\frac{2s^2 + 2u^2}{t^2}$	2007.07281v1 Eq. (3) source $\sigma_0 = \frac{\alpha\alpha_s}{sQ^2} \frac{2(\hat{s}^2 + \hat{u}^2)}{\hat{t}^2}.$	Strip electromagnetic coupling and quark charge factors; massless s+t+u=0. Preserve the scalar hard shape.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
B01 provenance. Fresh generated Born tensor; source-shared approved FeynCalc construction. Native generation rerun in this workflow. Historical group status: EXACT. Historical conversion: Strip the literal coupling and phase-space prefactor only for this kinematic comparison. Full literal comparison is B04. Archived v0.4.0 note: Kinematic factor only; N02/B04 retain the independent coupling discrepancy. New-minus-old: 0. New-minus-converted-reference: 0. B01 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
B02 Born UT / alpha^2	$\frac{2s^2}{t^2} - 2 + \frac{2u^2}{t^2}$	$\frac{2s^2 - 2t^2 + 2u^2}{t^2}$	2007.07281v1 Eq. (21) source $\sigma_0 \rightarrow \sigma_0^{\text{Collins}} = \frac{\alpha\alpha_s}{sQ^2} \frac{4\hat{s}\hat{u}}{-\hat{t}^2},$	Physical transverse spin interference; strip the same electromagnetic factors; s+t+u=0.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
B02 provenance. Fresh generated Born spin tensor; source-shared physical external-spin projector. Native generation rerun. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Use t=-s-u. Strip prefactor only for kinematic comparison. Full literal comparison is B05. New-minus-old: 0. New-minus-converted-reference: 0. B02 source/export bindings ; executable comparison .					
B03 Spin transfer D_NN / Born	$-\frac{2(y-1)}{y^2 - 2y + 2}$	$-\frac{2(y-1)}{y^2 - 2y + 2}$	1505.05589v1 Eq. (5) source $D_{NN} = \frac{2(1-y)}{1 + (1-y)^2}.$	Form the actual UT/UU quotient, then use t=-sy and u=-s(1-y), s>0, 0<y<1. This review-only quotient is not a separate production derivation.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
B03 provenance. Independent symbolic readback of the newly generated two Born tensor exports. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Use t=-s y,u=-s(1-y),0<y<1. Compare the kinematic ratio only; native and SIDIS harmonic definitions are separately mapped. New-minus-old: 0. New-minus-converted-reference: 0. B03 source/export bindings ; executable comparison .					
B04 Full Born UU prefactor	UNVERIFIED: no new expression mapped.	$\frac{\alpha_{\text{EM}}^2 (2s^2 + 2u^2)}{Q^2 St^2}$	2007.07281v1 Eq. (3) source $\sigma_0 = \frac{\alpha\alpha_s}{sQ^2} \frac{2(\hat{s}^2 + \hat{u}^2)}{\hat{t}^2}.$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
B04 provenance. Not established by the current new-route comparison record. Historical group status: MISMATCH. Historical conversion: Map paper alpha to native alphaEM and paper s to hadronicS; hatted invariants to s,t,u. Keep printed alpha_s untouched. Literal reference discrepancy; native normalization supported. Archived v0.4.0 note: Literal reference discrepancy; native normalization supported. No hidden alpha_s extraction or author-confirmed correction established. The formula above is the common Born normalization extracted from the bound full expression, not a replacement export. B04 source/export bindings ; executable comparison .					
B05 Full Born UT prefactor	UNVERIFIED: no new expression mapped.	$\frac{\alpha_{\text{EM}}^2 (2s^2 - 2t^2 + 2u^2)}{Q^2 St^2}$	2007.07281v1 Eq. (21) source $\sigma_0 \rightarrow \sigma_0^{\text{Collins}} = \frac{\alpha\alpha_s}{sQ^2} \frac{4\hat{s}\hat{u}}{-\hat{t}^2},$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
B05 provenance. Not established by the current new-route comparison record. Historical group status: MISMATCH. Historical conversion: Map alpha to alphaEM and use t=-s-u. Keep printed alpha_s untouched. Literal reference discrepancy; native normalization supported. Archived v0.4.0 note: Literal reference discrepancy; native normalization supported. No hidden alpha_s extraction or author-confirmed correction established. The formula above is the common Born normalization extracted from the bound full expression, not a replacement export. B05 source/export bindings ; executable comparison .					
B06 Phase-space Jacobian / Born	UNVERIFIED: no new expression mapped.	Sy	2007.07281v1 Eq. (4) source $x = \frac{p_T^e e^{y_e}}{\sqrt{s} - p_T^e e^{-y_e}}.$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
B06 cont.	Same new-route scope.	Same accepted comparison.	2007.07281v1 Eq. (8) source $y = 1 - \frac{p_T^e}{\sqrt{s}} e^{-y_e},$	Continuation; same explicit map.	Same target qualifications.
B06 provenance. Not established by the current new-route comparison record. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Take absolute inverse determinant on 0<x<1,0<y<1,S>0. Differential azimuth d^2pT=(1/2)d(pT^2)dphi, so its integral is pi d(pT^2). Archived v0.4.0 note: Kinematic map is an independently derived convention relation attached to the native Born variables; see N02 for the full measure. B06 source/export bindings ; executable comparison .					
B07 Mandelstam map / Born	UNVERIFIED: no new expression mapped.	$(Sx, -Sxy, Sx(y-1))$	2007.07281v1 Eq. (5) source $\hat{s} = xs,$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
B07 cont.	Same new-route scope.	Same accepted comparison.	2007.07281v1 Eq. (6) source $\hat{t} = -Q^2 = -\sqrt{s} p_T^e e^{y_e} = -x\sqrt{s} p_T^{\text{jet}} e^{-y_{\text{jet}}},$	Continuation; same explicit map.	Same target qualifications.

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
B07 cont.	Same new-route scope.	Same accepted comparison.	2007.07281v1 Eq. (7) source $\hat{u} = -x\sqrt{s}p_T^e e^{-y_e} = -\sqrt{s}p_T^{\text{jet}} e^{y_{\text{jet}}}$,	Continuation; same explicit map.	Same target qualifications.
B07 provenance. Not established by the current new-route comparison record. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Map hadronic s to hadronicS and use the separately derived x,y/lightcone relations. Born s+t+u=0; SIDIS virtual-photon invariants are distinct. Archived v0.4.0 note: The Mandelstam identity is symbolic. Detailed native-versus-paper differential-variable identification is qualified separately in B08/N01. B07 source/export bindings ; executable comparison .					
B08 Process and angular decomposition	UNVERIFIED: no new expression mapped.	$\frac{j\text{Theta} \cot(\text{theta.J})}{pJ}$	2007.07281v1 Eq. (14) source $e + p(\vec{s}_T) \rightarrow e + (\text{jet}(\vec{q}_T) h(z_h, \vec{j}_T)) + X$.	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
B08 cont.	Same new-route scope.	Same accepted comparison.	2007.07281v1 Eq. (15) source $\frac{d\sigma^h(\vec{s}_T)}{d\mathcal{PS} dz_h d^2\vec{j}_T^h} = F_{UU}^h + \sin(\phi_s - \phi_h) F_{UT}^{\sin(\phi_s - \phi_h)}$,	Continuation; same explicit map.	Same target qualifications.
B08 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Native process, anti-kt/E-scheme axis and scalar harmonic definitions are present. The paper beam-transverse fraction versus native axis/light-cone fraction is only a leading-power identification; literal measured-variable and angular equivalence remains unresolved (N01). B08 source/export bindings ; executable comparison .					

Hard, inclusive jet and soft functions

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
H01 Spacelike hard / a	$-C_F L_Q^2 - 3C_F L_Q - 8C_F + \frac{\pi^2 C_F}{6}$	$\frac{C_F (-6L (L + 3) - 48 + \pi^2)}{6}$	2007.07281v1 Eq. (A1) source $H_q(Q, \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[-\ln^2 \left(\frac{\mu^2}{Q^2} \right) - 3 \ln \left(\frac{\mu^2}{Q^2} \right) - 8 + \frac{\pi^2}{6} \right],$	LQ=log(mu^2/Q^2) maps explicitly to old L. H=1+a h1, a=alpha_s/(2pi), spacelike Q^2>0; local pole subtraction has no finite term.	Old: EXACT AFTER EXPLICIT CONVERSION; paper: EXACT AFTER EXPLICIT CONVERSION
H01 provenance. Fresh generated virtual current, actual audited Kira reduction, protected SubTropica bubble, separate UV/IR ledger and native r06 contraction. Historical group status: EXACT. Historical conversion: No change of convention is used for this comparison. New-minus-old: 0. New-minus-converted-reference: 0. H01 source/export bindings ; executable comparison .					
H02 JCC hard finite term / a	UNVERIFIED: no new expression mapped.	$-C_F (L_Q (L_Q + 3) + 8)$	Literal equation in continuation: 1505.05589v1 Eq. (A20)	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
1505.05589v1 Eq. (A20) source $H_{\text{collins}}^{(\text{DIS})JCC}(Q; \mu) = H_{UU}^{(\text{DIS})JCC}(Q; \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[3 \ln \frac{Q^2}{\mu^2} - \ln^2 \left(\frac{Q^2}{\mu^2} \right) - 8 \right].$					
H02 cont.	Same new-route scope.	Same accepted comparison.	Literal equation in continuation: 1303.2129v1 Eq. (48)	Continuation; same explicit map.	Same target qualifications.
1303.2129v1 Eq. (48) source $\mathcal{H}_{UU,T}^a(Q; \mu) = e_a^2 \left(1 + \frac{C_F \alpha_s}{\pi} \left[\frac{3}{2} \ln \left(\frac{Q^2}{\mu^2} \right) - \frac{1}{2} \ln^2 \left(\frac{Q^2}{\mu^2} \right) - 4 \right] \right).$					
H02 provenance. Not established by the current new-route comparison record. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: KPSY A20: remove tree 1 and extract a. BP48: remove charge squared and tree 1, multiply the alpha_s/pi coefficient by 2, and map ln(Q^2/mu^2)=-LQ. Independently transcribed target is -CF(LQ^2+3LQ+8). Native h_C11=h_ep-CF pi^2/6. Archived v0.4.0 note: H01 remains the ep hard coefficient. This hard relation alone does not certify every finite per-leg matching convention. H02 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
H03 Finite hard/operator scheme balance	UNVERIFIED: no new expression mapped.	$\frac{C_F \left(-6L_Q^2 - 18L_Q - 48 + \pi^2 \right)}{6}$	1505.05589v1 Eq. (139) source $H_{\text{SIDIS}}(Q, \mu = Q) = 1 + \frac{\alpha_s(Q)}{2\pi} C_F (-8) \text{ ,}$ $H_{e^+e^-}(Q, \mu = Q) = 1 + \frac{\alpha_s(Q)}{2\pi} C_F \left(\pi^2 - 8 \right) \text{ .}$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
H03 provenance. Not established by the current new-route comparison record. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: h_new=h_old-kB-kF; kB=kF=h_old/2 for absorption of the entire hard coefficient. Thus h_new+kB+kF=h_old. Archived v0.4.0 note: Internal balance is exact; it is not an independent published derivation of each finite HF endpoint. H03 source/export bindings ; executable comparison .					
H04 Inclusive quark jet / a	UNVERIFIED: no new expression mapped.	$\frac{C_F L_J^2}{2} + \frac{3C_F L_J}{2} - \frac{3\pi^2 C_F}{4} + \frac{13C_F}{2}$	2007.07281v1 Eq. (A2) source $J_q(p_T R, \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[\frac{1}{2} \ln^2 \left(\frac{\mu^2}{p_T^2 R^2} \right) + \frac{3}{2} \ln \left(\frac{\mu^2}{p_T^2 R^2} \right) + \frac{13}{2} - \frac{3}{4} \pi^2 \right] \text{ ,}$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
H04 provenance. Not established by the current new-route comparison record. Historical group status: EXACT. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Inclusive jet is compared here only. It is replaced by the fragmenting jet in d19, never multiplied beside it. H04 source/export bindings ; executable comparison .					
H05 Remaining measured soft / a	$2B_q C_F Y + B_q C_F r - \frac{C_F r^2}{2}$	$2B_q C_F Y + B_q C_F r - \frac{C_F r^2}{2}$	2007.07281v1 Eq. (A3) source $S_q(\vec{b}_T, y_{\text{jet}}, R, \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[-\ln \left(\frac{e^{-2y_{\text{jet}}}}{R^2} \right) \ln \left(\frac{\mu^2}{\mu_b^2} \right) - \frac{1}{2} \ln^2 \left(\frac{1}{R^2} \right) \right] \text{ ,}$	Bq=log(mu^2 gamma_E)/4), bq^2 exp(2 r=log R^2, Y=jet rapidity. Eta cancellation precedes epsilon expansion; the fixed local UV subtraction removes poles only.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
H05 provenance. Physical eikonal-current and conditional-angle derivation consumes the actual measured M111 sphere normalization and combines native global, cs_out and standard-soft nodes. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Bq=ln(mu^2 b_recoil^2/c0^2), r=ln R^2, Y=y_jet; eta-before-epsilon and the declared narrow-cone allocation. New-minus-old: 0. New-minus-converted-reference: 0. H05 source/export bindings ; executable comparison .					

Regulated soft constituents and subtractions

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
S01 Regulated global soft / a	New expression in continuation.	Accepted expression in continuation.	2007.07281v1 Eq. (A3) source $S_q(\vec{b}_T, y_{\text{jet}}, R, \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[-\ln \left(\frac{e^{-2y_{\text{jet}}}}{R^2} \right) \ln \left(\frac{\mu^2}{\mu_b^2} \right) - \frac{1}{2} \ln^2 \left(\frac{1}{R^2} \right) \right] \text{ ,}$	a=alpha_s/(2pi); eta expansion at fixed epsilon before epsilon expansion; Bq/Bh, V and g retain their declared scale definitions. Compare delta/rapidity poles and finite terms in the same regulated constituent basis.	Old: EXACT; paper: UNVERIFIED
S01 new expression: $B_q C_F V + 2B_q C_F Y - \frac{2B_q C_F}{\eta} + \frac{B_q C_F}{\epsilon} + \frac{C_F V}{\epsilon} + \frac{2C_F Y}{\epsilon} - \frac{\pi^2 C_F}{6} - \frac{2C_F}{\epsilon \eta} + \frac{2C_F}{\epsilon^2}$					
S01 accepted expression (archived): $-\frac{2B_q C_F}{\eta} + C_F \left(B_q (V + 2Y) - \frac{\pi^2}{6} \right) + \frac{C_F (B_q + V + 2Y)}{\epsilon} - \frac{2C_F}{\epsilon \eta} + \frac{2C_F}{\epsilon^2}$					
S01 provenance. Fresh physical soft-current/measure factor with actual native measured sphere. The paper does not separately print every regulated constituent; equality to the accepted expression is distinct from complete literature equality. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Fresh regulated constituent. The cited paper prints a renormalized combination or another regulator definition, not this individual eta-regulated Laurent expression; no symbolic literal equality is assigned. New-minus-old: 0. Complete literature difference is unverified. S01 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
S02 Regulated cs_out soft / a	New expression in continuation.	Accepted expression in continuation.	2007.07281v1 Eq. (A3) source $S_q(\vec{b}_T, y_{\text{jet}}, R, \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[-\ln\left(\frac{e^{-2y_{\text{jet}}}}{R^2}\right) \ln\left(\frac{\mu^2}{\mu_b^2}\right) - \frac{1}{2} \ln^2\left(\frac{1}{R^2}\right) \right],$	a=alpha_s/(2pi); eta expansion at fixed epsilon before epsilon expansion; Bq/Bh, V and g retain their declared scale definitions. Compare delta/rapidity poles and finite terms in the same regulated constituent basis.	Old: EXACT; paper: UNVERIFIED
S02 new expression: $-\frac{B_q^2 C_F}{2} + B_q C_F r - \frac{B_q C_F}{\epsilon} - \frac{C_F r^2}{2} + \frac{\pi^2 C_F}{12} + \frac{C_F r}{\epsilon} - \frac{C_F}{\epsilon^2}$					
S02 accepted expression (archived): $\frac{C_F \left(-6 (B_q - r)^2 + \pi^2 \right)}{12} + \frac{C_F (-B_q + r)}{\epsilon} - \frac{C_F}{\epsilon^2}$					
S02 provenance. Fresh physical soft-current/measure factor with actual native measured sphere. The paper does not separately print every regulated constituent; equality to the accepted expression is distinct from complete literature equality. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Fresh regulated constituent. The cited paper prints a renormalized combination or another regulator definition, not this individual eta-regulated Laurent expression; no symbolic literal equality is assigned. New-minus-old: 0. Complete literature difference is unverified. S02 source/export bindings; executable comparison .					
S03 Regulated standard soft / a	New expression in continuation.	Accepted expression in continuation.	Literal equation in continuation: 1303.2129v1 Eq. (4)	a=alpha_s/(2pi); eta expansion at fixed epsilon before epsilon expansion; Bq/Bh, V and g retain their declared scale definitions. Compare delta/rapidity poles and finite terms in the same regulated constituent basis.	Old: EXACT; paper: UNVERIFIED
S03 new expression: $-B_q^2 C_F + 2B_q C_F V - \frac{4B_q C_F}{\eta} + \frac{2C_F V}{\epsilon} - \frac{\pi^2 C_F}{6} - \frac{4C_F}{\epsilon \eta} + \frac{2C_F}{\epsilon^2}$					
S03 accepted expression (archived): $-\frac{4B_q C_F}{\eta} + \frac{2C_F V}{\epsilon} - \frac{C_F (6B_q (B_q - 2V) + \pi^2)}{6} - \frac{4C_F}{\epsilon \eta} + \frac{2C_F}{\epsilon^2}$					
1303.2129v1 Eq. (4) source $\tilde{f}_1(x, \mathbf{b}_T; \mu, \zeta_F) = \tilde{f}_1^{\text{unsub}}(x, \mathbf{b}_T; \mu; y_P - (-\infty)) \sqrt{\frac{\tilde{S}_{(0)}(\mathbf{b}_T; +\infty, y_s)}{\tilde{S}_{(0)}(\mathbf{b}_T; +\infty, -\infty) \tilde{S}_{(0)}(\mathbf{b}_T; y_s, -\infty)}} Z_F Z_2.$					
S03 provenance. Fresh physical soft-current/measure factor with actual native measured sphere. The paper does not separately print every regulated constituent; equality to the accepted expression is distinct from complete literature equality. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Fresh regulated constituent. The cited paper prints a renormalized combination or another regulator definition, not this individual eta-regulated Laurent expression; no symbolic literal equality is assigned. New-minus-old: 0. Complete literature difference is unverified. S03 source/export bindings; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
S04 Regulated injet soft / a	New expression in continuation.	Accepted expression in continuation.	2007.07281v1 Eq. (A3) source $S_q(\vec{b}_T, y_{\text{jet}}, R, \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[-\ln\left(\frac{e^{-2y_{\text{jet}}}}{R^2}\right) \ln\left(\frac{\mu^2}{\mu_b^2}\right) - \frac{1}{2} \ln^2\left(\frac{1}{R^2}\right) \right],$	a=alpha_s/(2pi); eta expansion at fixed epsilon before epsilon expansion; Bq/Bh, V and g retain their declared scale definitions. Compare delta/rapidity poles and finite terms in the same regulated constituent basis.	Old: EXACT; paper: UNVERIFIED
S04 new expression: $-\frac{B_h^2 C_F}{2} + B_h C_F V - B_h C_F g - \frac{2B_h C_F}{\eta} + \frac{C_F V}{\epsilon} - \frac{\pi^2 C_F}{12} - \frac{C_F g}{\epsilon} - \frac{2C_F}{\epsilon\eta} + \frac{C_F}{\epsilon^2}$					
S04 accepted expression (archived): $-\frac{2B_h C_F}{\eta} - \frac{C_F (6B_h (B_h - 2V + 2g) + \pi^2)}{12} + \frac{C_F (V - g)}{\epsilon} - \frac{2C_F}{\epsilon\eta} + \frac{C_F}{\epsilon^2}$					
S04 provenance. Fresh physical soft-current/measure factor with actual native measured sphere. The paper does not separately print every regulated constituent; equality to the accepted expression is distinct from complete literature equality. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Fresh regulated constituent. The cited paper prints a renormalized combination or another regulator definition, not this individual eta-regulated Laurent expression; no symbolic literal equality is assigned. New-minus-old: 0. Complete literature difference is unverified. S04 source/export bindings ; executable comparison .					
S05 Rapid- ity/UV tion / a	UNVERIFIED: no new expression mapped.	beam = 0 fragment = 0 soft = 0 jet = 0	No additional published counterpart.	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
S05 provenance. Not established by the current new-route comparison record. Historical group status: EXACT. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Explicit poles of fresh bare plus counterterm combinations vanish. This establishes the native subtraction algebra, not equality of individually regulated constituents to another regulator scheme. S05 source/export bindings ; executable comparison .					

Anomalous dimensions and scale relations

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
G01 gamma_mu hard / a	$-C_F (2L_Q + 3)$	$-C_F (2L_Q + 3)$	2007.07281v1 Eq. (A1) source $H_q(Q, \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[-\ln^2\left(\frac{\mu^2}{Q^2}\right) - 3\ln\left(\frac{\mu^2}{Q^2}\right) - 8 + \frac{\pi^2}{6} \right],$	d/dlog(mu^2)=d/dLQ, with a=alpha_s/(2pi) and running-coupling terms beyond this order omitted.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
G01 provenance. Review-only exact derivative of the actual freshly integrated virtual hard function. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Derivative with respect to ln(mu^2), not ln(mu); hold physical scales fixed. New-minus-old: 0. New-minus-converted-reference: 0. G01 source/export bindings ; executable comparison .					
G02 gamma_mu jet / a	UNVERIFIED: no new expression mapped.	$C_F \left(L_J + \frac{3}{2} \right)$	2007.07281v1 Eq. (A2) source $J_q(p_T R, \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[\frac{1}{2} \ln^2\left(\frac{\mu^2}{p_T^2 R^2}\right) + \frac{3}{2} \ln\left(\frac{\mu^2}{p_T^2 R^2}\right) + \frac{13}{2} - \frac{3}{4} \pi^2 \right],$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
G02 provenance. Not established by the current new-route comparison record. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Derivative with respect to ln(mu^2), not ln(mu); hold physical scales fixed. G02 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
G03 gamma_mu soft / a	$C_F (2Y + r)$	$C_F (2Y + r)$	2007.07281v1 Eq. (A3) source $S_q(\vec{b}_T, y_{\text{jet}}, R, \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[-\ln\left(\frac{e^{-2y_{\text{jet}}}}{R^2}\right) \ln\left(\frac{\mu^2}{\mu_b^2}\right) - \frac{1}{2} \ln^2\left(\frac{1}{R^2}\right) \right],$	d/dlog(mu^2)=d/dBq at fixed radius R and jet rapidity Y; a=alpha_s/(2pi).	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
G03 provenance. Review-only exact derivative of the actual measured recoil-soft finite function. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Derivative with respect to ln(mu^2), not ln(mu); hold physical scales fixed. New-minus-old: 0. New-minus-converted-reference: 0. G03 source/export bindings ; executable comparison .					
G04 gamma_mu beam_UU / a	$\frac{C_F (2T_q + 3)}{2}$	$C_F \left(T_q + \frac{3}{2}\right)$	Literal equation in continuation: 1505.05589v1 Eq. (A12)	Differentiate the actual finite delta coefficient at fixed physical zeta: d/dlog(mu^2)=d/dBq+d/dTq, then include the independently exported collinear delta evolution term. a=alpha_s/(2pi).	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
1505.05589v1 Eq. (A12) source					
$h_1^{sub(JCC)}(x_B, b; \zeta, \mu) \Big _{\alpha_s} = \frac{\alpha_s}{2\pi} \int_{x_B}^1 \frac{dx}{x} h_1(x, \bar{\mu}) \left\{ \ln \frac{c_0^2}{b^2 \bar{\mu}^2} P_{q \rightarrow q}^{h_1}(\xi) + C_F \delta(1 - \xi) \left[\frac{3}{2} \ln \frac{b^2 \mu^2}{c_0^2} + \frac{1}{2} \ln^2 \left(\frac{\zeta_c^2}{\mu^2} \right) - \frac{1}{2} \ln^2 \left(\frac{\zeta_c^2 b^2}{c_0^2} \right) \right] \right\}.$					
G04 provenance. Review-only exact derivative readback of generated matching and splitting exports; no separately generated anomalous-dimension packet is claimed. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: KPSY zeta_c^2 is native zeta. A12 delta coefficient is CF[3B/2+X^2/2-(X+B)^2/2], X=-T; d/dln(mu^2)=partial_B+partial_T gives CF(T+3/2). For A17, differentiate at both printed rapidity symbols fixed; its last squared log is mu independent. The missing c on its second hat-zeta is preserved, not repaired. Fragment T=T0-g. New-minus-old: 0. New-minus-converted-reference: 0. G04 source/export bindings ; executable comparison .					
G05 gamma_mu fragment_UU / a	$\frac{C_F (2Th + 3)}{2}$	$C_F \left(T_0 - g + \frac{3}{2}\right)$	Literal equation in continuation: 1505.05589v1 Eq. (A17)	Differentiate at fixed zetaJ, d/dlog(mu^2)=d/dBh+d/dTh, and include the collinear delta. Explicit comparison map Th=T0-g follows zetaJ=zeta0 exp(g).	Old: EXACT AFTER EXPLICIT CONVERSION; paper: EXACT AFTER EXPLICIT CONVERSION
1505.05589v1 Eq. (A17) source					
$\tilde{H}_1^{\perp \alpha sub(JCC)}(x_B, b; \zeta, \mu) \Big _{\alpha_s} = \frac{ib^\alpha}{2} \frac{\alpha_s}{2\pi} \int \frac{dz}{z} \hat{H}_{h/q}^{(3)}(z) \left\{ \ln \frac{c_0^2 \hat{\xi}^2}{b^2 \bar{\mu}^2} \hat{P}_{q \rightarrow q}^c(\hat{\xi}) + C_F \delta(1 - \hat{\xi}) \left[\frac{3}{2} \ln \frac{b^2 \mu^2}{c_0^2} + \frac{1}{2} \ln^2 \left(\frac{\hat{\xi}_c^2}{\mu^2} \right) - \frac{1}{2} \ln^2 \left(\frac{\hat{\xi}_c^2 b^2}{c_0^2} \right) \right] \right\}.$					
G05 provenance. Review-only exact scale derivative of native measured fragmentation; scale conventions are declared operator inputs. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: KPSY zeta_c^2 is native zeta. A12 delta coefficient is CF[3B/2+X^2/2-(X+B)^2/2], X=-T; d/dln(mu^2)=partial_B+partial_T gives CF(T+3/2). For A17, differentiate at both printed rapidity symbols fixed; its last squared log is mu independent. The missing c on its second hat-zeta is preserved, not repaired. Fragment T=T0-g. New-minus-old: 0. New-minus-converted-reference: 0. G05 source/export bindings ; executable comparison .					
G06 Spin-independent evolution / a	beam = 0 fragment = 0	beam = 0 fragment = 0	1303.2129v1 Eq. (21) source $\frac{d \ln \tilde{f}(x, \mathbf{b}_T; \mu, \zeta_F)}{d \ln \mu} = \gamma_F(g(\mu); \zeta_F/\mu^2).$	Compare transverse-spin and unpolarized scale derivatives in the same physical spin/Fourier continuation and at the same rapidity scale.	Old: EXACT; paper: UNVERIFIED
G06 provenance. Independent derivative comparison of the four actual measured delta coefficients and their collinear endpoint terms. Historical group status: EXACT. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Spin-independent evolution of the generated operators; finite spin matching is a different comparison. New-minus-old: 0. Complete literature difference is unverified. G06 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
G07 Soper a	Collins- kernels / beam = $-2B_q C_F$ fragment = $-2B_h C_F$	beam = $-2B_q C_F$ fragment = $-2B_h C_F$	Literal equation in continuation: 1505.05589v1 Eq. (A12)	The displayed Collins-Soper kernel uses d/dlog(sqrt(zeta)); hence -2 d/dT of the actual finite coefficient. Its d/dlog(zeta) counterpart is half this expression.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
	1505.05589v1 Eq. (A12) source $h_1^{sub(JCC)}(x_B, b; \zeta, \mu) \Big _{\alpha_s} = \frac{\alpha_s}{2\pi} \int_{x_B}^1 \frac{dx}{x} h_1(x, \bar{\mu}) \left\{ \ln \frac{c_0^2}{b^2 \bar{\mu}^2} P_{q \rightarrow q}^{h_1}(\xi) + C_F \delta(1 - \xi) \left[\frac{3}{2} \ln \frac{b^2 \mu^2}{c_0^2} + \frac{1}{2} \ln^2 \left(\frac{\zeta_c^2}{\mu^2} \right) - \frac{1}{2} \ln^2 \left(\frac{\zeta_c^2 b^2}{c_0^2} \right) \right] \right\}.$				
G07 cont.	Same new-route scope.	Same accepted comparison.	1505.05589v1 Eq. (22) source $\frac{\partial \ln \tilde{f}_q(x_B, b; \zeta, \mu)}{\partial \ln \sqrt{\zeta_F}} = \frac{\partial \ln \tilde{D}_q(z_h, b, \zeta_D; \mu)}{\partial \ln \sqrt{\zeta_D}} = \tilde{K}(b, \mu),$	Continuation; same explicit map.	Same target qualifications.
G07 provenance. Review-only exact rapidity derivative of native measured coefficients, with explicit derivative normalization. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: From independently transcribed A12 delta bracket CF[3B/2+X ² /2-(X+B) ² /2], X=-T, -2 partial_T gives -2CF B. Native zeta=zeta_c ² . KPSY22 supplies the common PDF/FF CS kernel with independent Bq and Bh; no silent hat-zeta repair in A17. New-minus-old: 0. New-minus-converted-reference: 0. G07 source/export bindings ; executable comparison .					
G08 CS/mu inte- grability / a	0	0	1303.2129v1 Eq. (18) source $\frac{\partial \ln \tilde{f}(x, \mathbf{b}_T; \mu, \zeta_F)}{\partial \ln \sqrt{\zeta_F}} = \tilde{K}(\mathbf{b}_T; \mu)$	Check dK/dlog(mu^2)+2 d gamma_mu/dTq=0 with K defined by d/dlog(sqrt(zeta)).	Old: EXACT; paper: UNVERIFIED
	Same new-route scope.	Same accepted comparison.	1303.2129v1 Eq. (20) source $\frac{d\tilde{K}(\mathbf{b}_T; \mu)}{d \ln \mu} = -\gamma_K(g(\mu))$	Continuation; same explicit map.	Same target qualifications.
G08 provenance. Exact mixed-derivative check of generated finite matching and collinear evolution; no floating-point inference. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Mixed rapidity/mu derivatives in the same log conventions. New-minus-old: 0. Complete literature difference is unverified. G08 source/export bindings ; executable comparison .					
G09 UU/UT scale cancellation / a	UNVERIFIED: no new expression mapped.	Accepted expression in continuation.	Literal equation in continuation: 2007.07281v1 Eq. (17)	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
G09 accepted expression (archived): $-2C_F L_Q + C_F T_0 + 2C_F Y - C_F g + C_F r + C_F (2L_Q - T_0 - 2Y + g - r)$					
2007.07281v1 Eq. (17) source $F_{UU}^h = \sigma_0 H_q(Q, \mu) \sum_q e_q^2 \mathcal{G}_q^h(z_h, \vec{j}_T, p_T^{\text{jet}} R, \mu) \times \int \frac{d^2 \vec{b}_T}{(2\pi)^2} e^{i\vec{q}_T \cdot \vec{b}_T} f_q^{\text{TMD}}(x, \vec{b}_T, \mu) S_q(\vec{b}_T, y_{\text{jet}}, R, \mu).$					
G09 provenance. Not established by the current new-route comparison record. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: zeta_P zeta_J=Q ⁴ exp(2Y) R ² , with the common collinear scale. Full HF matching-scale consistency is separately C15/E03. G09 source/export bindings ; executable comparison .					
G10 Beam rapid- ity choice vs Ap- pendix Q	UNVERIFIED: no new expression mapped.	beam_shift = $-B_q C_F \chi$ soft_shift = $B_q C_F \chi$	2007.07281v1 Eq. (A16) source $f_q^{\text{TMD}}(x_B, \vec{b}_T, Q) = f_q^{\text{TMD}}(x_B, \vec{b}_T, \mu_b) \times e^{-\frac{1}{2} S_{\text{pert}}(Q; b_*) - S_{\text{NP}}^f(Q; b_T)}$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
	Same new-route scope.	Same accepted comparison.	1303.2129v1 Eq. (41) source $\mu = Q, \quad \mu_b = 2e^{-\gamma_E}/b_* \equiv b_0/b_*, \quad \zeta_F = Q^2, \quad \zeta_{F0} = Q_0^2.$	Continuation; same explicit map.	Same target qualifications.
G10 provenance. Not established by the current new-route comparison record. Historical group status: EXACT. Historical conversion: The actual d14 common rapidity shift is compensated between beam and soft; the paper Q ² abbreviation is not silently equated to native zeta_P. Archived v0.4.0 note: Exact compensating identity; literal measurement/axis qualifications remain. G10 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
G11 Fragment- ing jet replaces inclusive J	UNVERIFIED: no new expression mapped.	0	2007.07281v1 Eq. (12) source $F_{UU} = \sigma_0 H_q(Q, \mu) \sum_q e_q^2 J_q(p_T^{\text{jet}} R, \mu)$ $\times \int \frac{d^2 \vec{b}_T}{(2\pi)^2} e^{i\vec{q}_T \cdot \vec{b}_T} f_q^{\text{TMD}}(x, \vec{b}_T, \mu) S_q(\vec{b}_T, y_{\text{jet}}, R, \mu),$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
G11 cont.	Same new-route scope.	Same accepted comparison.	Literal equation in continuation: 2007.07281v1 Eq. (17)	Continuation; same explicit map.	Same target qualifications.
2007.07281v1 Eq. (17) source $F_{UU}^h = \sigma_0 H_q(Q, \mu) \sum_q e_q^2 \mathcal{G}_q^h(z_h, \vec{j}_T, p_T^{\text{jet}} R, \mu)$ $\times \int \frac{d^2 \vec{b}_T}{(2\pi)^2} e^{i\vec{q}_T \cdot \vec{b}_T} f_q^{\text{TMD}}(x, \vec{b}_T, \mu) S_q(\vec{b}_T, y_{\text{jet}}, R, \mu).$					
G11 provenance. Not established by the current new-route comparison record. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: zeta_J=pT ² R ² ; fragmenting jet replaces inclusive J. d18 is an additional conversion after intrinsic matching. G11 source/export bindings ; executable comparison .					

Splitting kernels and distributions

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
P01 P_qq / LO	$\text{delta} = \frac{3C_F}{2}$ $\text{D0} = 2C_F$ $\text{D1} = 0$ $\text{regular} = -C_F z - C_F$	$\text{delta} = \frac{3C_F}{2}$ $\text{D0} = 2C_F$ $\text{D1} = 0$ $\text{regular} = C_F(-z - 1)$	2007.07281v1 Eq. (A10) source $P_{q \leftarrow q}(z) = C_F \left[\frac{1 + z^2}{(1 - z)_+} + \frac{3}{2} \delta(1 - z) \right],$	Distribution basis delta(1-z), [1/(1-z)]+, [log(1-z)/(1-z)]+ and regular coefficient on 0<z<1; a=alpha_s/(2pi). Endpoint delta and plus are compared separately.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
P01 cont.	Same new-route scope.	Same accepted comparison.	1505.05589v1 Eq. (43) source $P_{q \leftarrow q}(z) = C_F \left[\frac{1 + z^2}{(1 - z)_+} + \frac{3}{2} \delta(1 - z) \right],$	Continuation; same explicit map.	Same target qualifications.
P01 provenance. Native measured real D-dimensional numerator with explicit collinear endpoint completion; the four exported distribution coefficients are read independently. Historical group status: UNVERIFIED. Historical conversion: Canonical distribution basis: delta, D0=[1/(1-z)]+, D1, regular. Move numerator-dependent plus terms into regular terms explicitly. New-minus-old: 0. New-minus-converted-reference: 0. Shared equation IDs: P01, P05. P01 source/export bindings ; executable comparison .					
P02 P_gq / LO	$\text{delta} = 0$ $\text{D0} = 0$ $\text{D1} = 0$ $\text{regular} = C_F z - 2C_F + \frac{2C_F}{z}$	$\text{delta} = 0$ $\text{D0} = 0$ $\text{D1} = 0$ $\text{regular} = \frac{C_F(z^2 - 2z + 2)}{z}$	2007.07281v1 Eq. (A11) source $P_{g \leftarrow q}(z) = C_F \frac{1 + (1 - z)^2}{z}.$	Distribution basis delta(1-z), [1/(1-z)]+, [log(1-z)/(1-z)]+ and regular coefficient on 0<z<1; a=alpha_s/(2pi). Endpoint delta and plus are compared separately.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
P02 cont.	Same new-route scope.	Same accepted comparison.	1505.05589v1 Eq. (44) source $P_{g \leftarrow q}(z) = C_F \frac{1 + (1 - z)^2}{z}.$	Continuation; same explicit map.	Same target qualifications.
P02 provenance. Native measured real D-dimensional numerator with explicit collinear endpoint completion; the four exported distribution coefficients are read independently. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Canonical distribution basis: delta, D0=[1/(1-z)]+, D1, regular. Move numerator-dependent plus terms into regular terms explicitly. New-minus-old: 0. New-minus-converted-reference: 0. P02 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
P03 P_qg / LO	delta = 0	delta = 0	Literal equation in continuation: 1303.2129v1 Eq. (35)	Distribution basis delta(1-z), [1/(1-z)]+, [log(1-z)/(1-z)]+ and regular coefficient on 0<z<1; a=alpha_s/(2pi). Endpoint delta and plus are compared separately.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
	D0 = 0	D0 = 0			
	D1 = 0	D1 = 0			
	regular = 2T_R z^2 - 2T_R z + T_R	regular = T_R (2z^2 - 2z + 1)			
1303.2129v1 Eq. (35) source $\tilde{C}_{g/j}(x, \boldsymbol{b}_T; \mu, \zeta_F/\mu^2) = \frac{\alpha_s T_f}{\pi} \left\{ \ln \left(\frac{2e^{-\gamma_E}}{\mu b_T} \right) (x^2 + (1-x)^2) + x(1-x) \right\} + \mathcal{O}(\alpha_s^2).$					
P03 provenance. Native measured real D-dimensional numerator with explicit collinear endpoint completion; the four exported distribution coefficients are read independently. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: BP35: alpha_s Tf/pi ln[c0/(mu b)]=a TR ln[c0^2/(mu^2 b^2)], Tf=TR, x=z. Minus the coefficient of B=ln(mu^2 b^2/c0^2) is P_qg. Canonical distribution basis: delta, D0=[1/(1-z)]+, D1, regular. Move numerator-dependent plus terms into regular terms explicitly. New-minus-old: 0. New-minus-converted-reference: 0. P03 source/export bindings ; executable comparison .					
P04 P_transversity / LO	delta = 3C_F/2	delta = 3C_F/2	1505.05589v1 Eq. (64) source $\hat{P}_{q \leftarrow q}^c(z) = C_F \left[\frac{2z}{(1-z)_+} + \frac{3}{2} \delta(1-z) \right].$	Distribution basis delta(1-z), [1/(1-z)]+, [log(1-z)/(1-z)]+ and regular coefficient on 0<z<1; a=alpha_s/(2pi). Endpoint delta and plus are compared separately.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
	D0 = 2C_F	D0 = 2C_F			
	D1 = 0	D1 = 0			
	regular = -2C_F	regular = -2C_F			
P04 cont.	Same new-route scope.	Same accepted comparison.	1505.05589v1 Eq. (70) source $P_{q \rightarrow q}^{h_1}(\hat{x}) = C_F \left[\frac{2\hat{x}}{(1-\hat{x})_+} + \frac{3}{2} \delta(1-\hat{x}) \right].$	Continuation; same explicit map.	Same target qualifications.
P04 provenance. Native measured real D-dimensional numerator with explicit collinear endpoint completion; the four exported distribution coefficients are read independently. Historical group status: UNVERIFIED. Historical conversion: Canonical distribution basis: delta, D0=[1/(1-z)]+, D1, regular. Move numerator-dependent plus terms into regular terms explicitly. New-minus-old: 0. New-minus-converted-reference: 0. Shared equation IDs: P04, P06, E01, E02. P04 source/export bindings ; executable comparison .					
P07 Dimensional real kernels / d=4-2eps	New expression in continuation.	qq = C_F (epsilon z^2 - 2 epsilon z + epsilon - z^2 - 1) / (z - 1)	2007.07281v1 Eq. (A10) source $P_{q \leftarrow q}(z) = C_F \left[\frac{1 + z^2}{(1-z)_+} + \frac{3}{2} \delta(1-z) \right],$	D=4-2epsilon. Spacelike beam qq/qg and final-state gq numerators use their actual spin/color normalizations. The four-dimensional printed splitting functions do not establish their epsilon-dependent completion.	Old: EXACT; paper: UNVERIFIED
		qg = T_R (epsilon - 2 z^2 + 2 z - 1) / (epsilon - 1)			
		gq = -C_F (epsilon z^2 - z^2 + 2 z - 2) / z			
P07 new expression: qq = C_F (epsilon z^2 - 2 epsilon z + epsilon - z^2 - 1) / (z - 1) qg = T_R (epsilon - 2 z^2 + 2 z - 1) / (epsilon - 1) gq = -C_F (epsilon z^2 - z^2 + 2 z - 2) / z					
P07 cont.	Same new-route scope.	Same accepted comparison.	2007.07281v1 Eq. (A11) source $P_{g \leftarrow q}(z) = C_F \frac{1 + (1-z)^2}{z}.$	Continuation; same explicit map.	Same target qualifications.
P07 provenance. Fresh native generating traces, independently read from final r00 measured_tmd_operators.wl by the focused projection exporter; no paper target supplied these numerators. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: The fresh D-dimensional numerators retain epsilon terms. These reference equations specify four-dimensional kernels; they do not independently determine those terms. New-minus-old: 0. Complete literature difference is unverified. P07 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
P08 Independent distribution moments	qq = 0	qq = 0	No additional published counterpart.	Independent test-function action on phi(z)=1 over [0,1]. D0 and D1 plus terms annihilate the constant test; retain the actual delta plus regular integral.	Old: EXACT; paper: UNVERIFIED
	transversity = $-\frac{C_F}{2}$	transversity = $-\frac{C_F}{2}$			
P08 provenance. Review-only exact distribution moment of the newly generated four-component splitting exports, not a floating-point moment test. Historical group status: EXACT. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: The old numerical-only moment evidence remains in the shared history. This new calculation explicitly integrates the fresh canonical distribution. New-minus-old: 0. Complete literature difference is unverified. P08 source/export bindings ; executable comparison .					

Finite matching and scheme allocation

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
M01 Finite f quark / a	delta = 0	delta = 0	2007.07281v1 Eq. (A6) source $C_{q\leftarrow q'}(x,\mu_b)=\delta_{q'q}\Big[\delta(1-x)+\frac{\alpha_s}{\pi}\left(\frac{C_F}{2}(1-x)\right)\Big],$	Canonical C11 comparison convention: set the declared natural logarithms to zero and apply the explicit b-independent finite operator conversion. The actual matching distribution is read as delta/plus/regular coefficients; fragmentation retains its z scaling.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
	D0 = 0	D0 = 0			
	D1 = 0	D1 = 0			
	regular = $-C_F z + C_F$	regular = $-C_F (z - 1)$			
M01 provenance. Generated measured operator numerator contracted with actual Kira M111 and protected native sphere, followed by explicit Fourier continuation and fixed subtraction/convention maps. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: ell in the fresh transport means ln(z); a=alpha_s/(2 pi), canonical KPSY_C11 boundary mu=mu_b and zeta=mu_b^2. Tree delta is separate. Log(z) times a plus kernel becomes an integrable ordinary function. Archived v0.4.0 note: This expression now comes from fresh d13 operator/amplitude and transverse-integral derivation. The historical Python finite formula is not used as its source. New-minus-old: 0. New-minus-converted-reference: 0. M01 source/export bindings ; executable comparison .					
M02 Finite f g / a	delta = 0	delta = 0	2007.07281v1 Eq. (A7) source $C_{q\leftarrow g}(x,\mu_b)=\frac{\alpha_s}{\pi}T_R x(1-x),$	Canonical C11 comparison convention: set the declared natural logarithms to zero and apply the explicit b-independent finite operator conversion. The actual matching distribution is read as delta/plus/regular coefficients; fragmentation retains its z scaling.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
	D0 = 0	D0 = 0			
	D1 = 0	D1 = 0			
	regular = $-2T_R z^2 + 2T_R z$	regular = $-2T_R z (z - 1)$			
M02 provenance. Generated measured operator numerator contracted with actual Kira M111 and protected native sphere, followed by explicit Fourier continuation and fixed subtraction/convention maps. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: ell in the fresh transport means ln(z); a=alpha_s/(2 pi), canonical KPSY_C11 boundary mu=mu_b and zeta=mu_b^2. Tree delta is separate. Log(z) times a plus kernel becomes an integrable ordinary function. Archived v0.4.0 note: This expression now comes from fresh d13 operator/amplitude and transverse-integral derivation. The historical Python finite formula is not used as its source. New-minus-old: 0. New-minus-converted-reference: 0. M02 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
M03 Finite d quark / a	New expression in continuation.	delta = 0	2007.07281v1 Eq. (A8) source $\hat{C}_{q'\leftarrow q}(z,\mu_b)=\delta_{q'q}\Big[\delta(1-z)+\frac{\alpha_s}{\pi}\Big(\frac{C_F}{2}(1-z)+P_{q\leftarrow q}(z)\ln z\Big)\Big],$	Canonical C11 comparison convention: Old: EXACT; set the declared natural logarithms paper: EXACT to zero and apply the explicit b- AFTER independent finite operator conversion. EXPLICIT The actual matching distribution is CONVERSION read as delta/plus/regular coefficients; fragmentation retains its z scaling.	
		D0 = 0			
		D1 = 0			
regular = $-\frac{C_F\left(2z^2\log\left(z\right)+z^2-2z+2\log\left(z\right)+1\right)}{z-1}$					
M03 new expression:					
delta = 0					
D0 = 0					
D1 = 0					
regular = $-\frac{2C_Fz^2\log\left(z\right)}{z-1}-\frac{C_Fz^2}{z-1}+\frac{2C_Fz}{z-1}-\frac{2C_F\log\left(z\right)}{z-1}-\frac{C_F}{z-1}$					
M03 provenance. Generated measured operator numerator contracted with actual Kira M111 and protected native sphere, followed by explicit Fourier continuation and fixed subtraction/convention maps. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: ell in the fresh transport means ln(z); a=alpha_s/(2 pi), canonical KPSY_C11 boundary mu=mu_b and zeta=mu_b^2. Tree delta is separate. Log(z) times a plus kernel becomes an integrable ordinary function. Archived v0.4.0 note: This expression now comes from fresh d13 operator/amplitude and transverse-integral derivation. The historical Python finite formula is not used as its source. New-minus-old: 0. New-minus-converted-reference: 0. M03 source/export bindings ; executable comparison .					
M04 Finite frag-mentation g from q / a	New expression in continuation.	delta = 0	2007.07281v1 Eq. (A9) source $\hat{C}_{g\leftarrow q}(z,\mu_b)=\frac{\alpha_s}{\pi}\left(\frac{C_F}{2}z+P_{g\leftarrow q}(z)\ln z\right).$	Canonical C11 comparison convention: Old: EXACT; set the declared natural logarithms paper: EXACT to zero and apply the explicit b- AFTER independent finite operator conversion. EXPLICIT The actual matching distribution is CONVERSION read as delta/plus/regular coefficients; fragmentation retains its z scaling.	
		D0 = 0			
		D1 = 0			
regular = $\frac{C_F\left(2z^2\log\left(z\right)+z^2-4z\log\left(z\right)+4\log\left(z\right)\right)}{z}$					
M04 new expression:					
delta = 0					
D0 = 0					
D1 = 0					
regular = $2C_Fz\log\left(z\right)+C_Fz-4C_F\log\left(z\right)+\frac{4C_F\log\left(z\right)}{z}$					
M04 provenance. Generated measured operator numerator contracted with actual Kira M111 and protected native sphere, followed by explicit Fourier continuation and fixed subtraction/convention maps. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: ell in the fresh transport means ln(z); a=alpha_s/(2 pi), canonical KPSY_C11 boundary mu=mu_b and zeta=mu_b^2. Tree delta is separate. Log(z) times a plus kernel becomes an integrable ordinary function. Archived v0.4.0 note: This expression now comes from fresh d13 operator/amplitude and transverse-integral derivation. The historical Python finite formula is not used as its source. New-minus-old: 0. New-minus-converted-reference: 0. M04 source/export bindings ; executable comparison .					
M05 Finite t quark / a		delta = 0	1505.05589v1 Eq. (62) source $\delta C_{q\leftarrow q'}(x,\mu_b)=\delta_{q'q}\Big[\delta(1-x)+\mathcal{O}(\alpha_s^2)\Big],$	Canonical C11 comparison convention: Old: EXACT; set the declared natural logarithms paper: EXACT to zero and apply the explicit b- AFTER independent finite operator conversion. EXPLICIT The actual matching distribution is CONVERSION read as delta/plus/regular coefficients; fragmentation retains its z scaling.	
		D0 = 0			
		D1 = 0			
regular = 0					
M05 provenance. Generated measured operator numerator contracted with actual Kira M111 and protected native sphere, followed by explicit Fourier continuation and fixed subtraction/convention maps. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: ell in the fresh transport means ln(z); a=alpha_s/(2 pi), canonical KPSY_C11 boundary mu=mu_b and zeta=mu_b^2. Tree delta is separate. Log(z) times a plus kernel becomes an integrable ordinary function. Archived v0.4.0 note: This expression now comes from fresh d13 operator/amplitude and transverse-integral derivation. The historical Python finite formula is not used as its source. New-minus-old: 0. New-minus-converted-reference: 0. M05 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
M06 Finite quark / a	delta = 0	delta = 0	1505.05589v1 Eq. (63) source $\delta\hat{C}_{q'\leftarrow q}^{(\text{SIDIS})}(z,\mu_b) = \delta_{q'q} \left[\delta(1-z) + \frac{\alpha_s}{\pi} \left(\hat{P}_{q\leftarrow q}^c(z) \ln z \right) \right] ,$	Canonical C11 comparison convention: Old: EXACT; set the declared natural logarithms paper: EXACT to zero and apply the explicit b-independent finite operator conversion. EXPLICIT The actual matching distribution is CONVERSION read as delta/plus/regular coefficients; fragmentation retains its z scaling.	
	D0 = 0	D0 = 0			
	D1 = 0	D1 = 0			
	regular = $-\frac{4C_F z \log(z)}{z-1}$	regular = $-\frac{4C_F z \log(z)}{z-1}$			
	M06 provenance. Generated measured operator numerator contracted with actual Kira M111 and protected native sphere, followed by explicit Fourier continuation and fixed subtraction/convention maps. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: ell in the fresh transport means ln(z); a=alpha_s/(2 pi), canonical KPSY_C11 boundary mu=mu_b and zeta=mu_b^2. Tree delta is separate. Log(z) times a plus kernel becomes an integrable ordinary function. Archived v0.4.0 note: This expression now comes from fresh d13 operator/amplitude and transverse-integral derivation. The historical Python finite formula is not used as its source. New-minus-old: 0. New-minus-converted-reference: 0. M06 source/export bindings ; executable comparison .				
M07 Finite larized OPE at arbitrary scales	New expression in continuation.	Accepted expression in continuation.	Literal equation in continuation: 1303.2129v1 Eq. (30)	For comparison only, independently identify Bq or Bh with B and Tq or Th with T within each channel; set ell=log z in the accepted generic-scale basis. Physical recoil and fragmentation Fourier variables remain distinct in the generated expression.	Old: EXACT AFTER EXPLICIT CONVERSION; paper: UNVERIFIED
M07 new expression: delta = $-\frac{B_q^2 C_F}{2} + B_q C_F T_q - \frac{\pi^2 C_F}{12}$ fqq : D0 = $-2B_q C_F$ D1 = 0 regular = $B_q C_F z + B_q C_F - C_F z + C_F$					
M07 accepted expression (archived): delta = $-\frac{B^2 C_F}{2} + B C_F T - \frac{\pi^2 C_F}{12}$ fqq : D0 = $-2B C_F$ D1 = 0 regular = $B C_F z + B C_F - C_F z + C_F$					
1303.2129v1 Eq. (30) source $\delta\tilde{C}_{j'/j}(x,\mathbf{b}_T;\mu;\zeta_F/\mu^2) = \delta_{j'j}\delta(1-x) + \delta_{j'j}\frac{\alpha_s C_F}{\pi}\left\{\ln\left(\frac{2e^{-\gamma_E}}{\mu b_T}\right)\left(\frac{2x}{1-x}\right)_+ + \right. \\ \left. + \delta(1-x)\left[-\ln^2\left(\frac{2e^{-\gamma_E}}{\mu b_T}\right) + \ln\left(\frac{2e^{-\gamma_E}}{\mu b_T}\right)\ln\left(\frac{\zeta_F}{\mu^2}\right)\right]\right\} + \mathcal{O}(\alpha_s^2) .$					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
M07 cont.	New expression in continuation.	Accepted expression in continuation.	Literal equation in continuation: 1505.05589v1 Eq. (A12)	Continuation; same explicit map.	Same target qualifications.
M07 new expression:					
fqg :	delta = 0				
	D0 = 0				
	D1 = 0				
	regular = $-2B_qT_Rz^2 + 2B_qT_Rz - B_qT_R - 2T_Rz^2 + 2T_Rz$				
M07 accepted expression (archived):					
fqg :	delta = 0				
	D0 = 0				
	D1 = 0				
	regular = $-2BT_Rz^2 + 2BT_Rz - BT_R - 2T_Rz^2 + 2T_Rz$				
1505.05589v1 Eq. (A12) source					
	$h_1^{sub(JCC)}(x_B, b; \zeta, \mu) \Big _{\alpha_s} = \frac{\alpha_s}{2\pi} \int_{x_B}^1 \frac{dx}{x} h_1(x, \bar{\mu}) \left\{ \ln \frac{c_0^2}{b^2 \bar{\mu}^2} P_{q \rightarrow q}^{h_1}(\xi) \right. \\ \left. + C_F \delta(1 - \xi) \left[\frac{3}{2} \ln \frac{b^2 \mu^2}{c_0^2} + \frac{1}{2} \ln^2 \left(\frac{\zeta_c^2}{\mu^2} \right) - \frac{1}{2} \ln^2 \left(\frac{\zeta_c^2 b^2}{c_0^2} \right) \right] \right\} .$				
M07 cont.	New expression in continuation.	Accepted expression in continuation.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
M07 new expression:					
dqg :	delta = $-\frac{B_h^2 C_F}{2} + B_h C_F T h - \frac{\pi^2 C_F}{12}$				
	D0 = $-2B_h C_F$				
	D1 = 0				
	regular = $\frac{B_h C_F z^2}{z - 1} - \frac{B_h C_F}{z - 1} - \frac{2 C_F z^2 \log(z)}{z - 1} - \frac{C_F z^2}{z - 1} + \frac{2 C_F z}{z - 1} - \frac{2 C_F \log(z)}{z - 1} - \frac{C_F}{z - 1}$				
M07 accepted expression (archived):					
dqg :	delta = $-\frac{B^2 C_F}{2} + B C_F T - \frac{\pi^2 C_F}{12}$				
	D0 = $-2B C_F$				
	D1 = 0				
	regular = $B C_F z + B C_F - \frac{2 C_F \ell z^2}{z - 1} - \frac{2 C_F \ell}{z - 1} - \frac{C_F z^2}{z - 1} + \frac{2 C_F z}{z - 1} - \frac{C_F}{z - 1}$				
M07 cont.	New expression in continuation.	Accepted expression in continuation.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
M07 new expression:					
dgq :	delta = 0				
	D0 = 0				
	D1 = 0				
	regular = $-B_h C_F z + 2B_h C_F - \frac{2B_h C_F}{z} + 2C_F z \log(z) + C_F z - 4C_F \log(z) + \frac{4C_F \log(z)}{z}$				
M07 accepted expression (archived):					
dgq :	delta = 0				
	D0 = 0				
	D1 = 0				
	regular = $-B C_F z + 2B C_F - \frac{2B C_F}{z} + 2C_F \ell z - 4C_F \ell + \frac{4C_F \ell}{z} + C_F z$				

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
M07 cont.	New expression in continuation.	Accepted expression in continuation.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
M07 new expression: $\text{delta} = -\frac{B_q^2 C_F}{2} + B_q C_F T_q - \frac{\pi^2 C_F}{12}$ hqq : $\text{D0} = -2B_q C_F$ $\text{D1} = 0$ $\text{regular} = 2B_q C_F$					
M07 accepted expression (archived): $\text{delta} = -\frac{B^2 C_F}{2} + B C_F T - \frac{\pi^2 C_F}{12}$ hqq : $\text{D0} = -2B C_F$ $\text{D1} = 0$ $\text{regular} = 2B C_F$					
M07 cont.	New expression in continuation.	Accepted expression in continuation.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
M07 new expression: $\text{delta} = -\frac{B_h^2 C_F}{2} + B_h C_F T h - \frac{\pi^2 C_F}{12}$ collins : $\text{D0} = -2B_h C_F$ $\text{D1} = 0$ $\text{regular} = \frac{2B_h C_F z}{z-1} - \frac{2B_h C_F}{z-1} - \frac{4C_F z \log(z)}{z-1}$					
M07 accepted expression (archived): $\text{delta} = -\frac{B^2 C_F}{2} + B C_F T - \frac{\pi^2 C_F}{12}$ collins : $\text{D0} = -2B C_F$ $\text{D1} = 0$ $\text{regular} = 2B C_F - \frac{4C_F \ell z}{z-1}$					
M07 provenance. All six native full-scale measured distributions; common angular/Fourier continuation and explicit operator normalization remain stated. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Fresh arbitrary-scale homogeneous distributions obtained from current canonical matching and collinear/TMD evolution, compared to the actual local OPE by the bound native conversion. Literal BP polarized endpoint/scheme identification still needs the complete per-reference map; no published symbolic equality is inferred from internal RG identities. New-minus-old: 0. Complete literature difference is unverified. M07 source/export bindings ; executable comparison .					
M08 Helicity and other matching	UNVERIFIED: no new expression polarized mapped.	No fresh analytical expression exported	Literal equation in continuation: 1303.2129v1 Eq. (29)	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
1303.2129v1 Eq. (29) source $\Delta \tilde{C}_{j'/j}(x, \mathbf{b}_T; \mu; \zeta_F/\mu^2) = \delta_{j'j} \delta(1-x) + \delta_{j'j} \frac{\alpha_s C_F}{\pi} \left\{ \ln \left(\frac{2e^{-\gamma_E}}{\mu b_T} \right) \left(\frac{1+x^2}{1-x} \right)_+ + \frac{1}{2}(1-x) + \right.$ $\left. + \delta(1-x) \left[-\ln^2 \left(\frac{2e^{-\gamma_E}}{\mu b_T} \right) + \ln \left(\frac{2e^{-\gamma_E}}{\mu b_T} \right) \ln \left(\frac{\zeta_F}{\mu^2} \right) \right] \right\} + \mathcal{O}(\alpha_s^2),$					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
M08 cont.	Same new-route scope.	Same accepted comparison.	Literal equation in continuation: 1303.2129v1 Eq. (36)	Continuation; same explicit map.	Same target qualifications.
1303.2129v1 Eq. (36) source $\Delta\tilde{C}_{g/j}(x, \mathbf{b}_T; \mu, \zeta_F/\mu^2) = \frac{\alpha_s T_f}{\pi} \left\{ \ln \left(\frac{2e^{-\gamma_E}}{\mu b_T} \right) (2x - 1) + (1 - x) \right\} + \mathcal{O}(\alpha_s^2).$					
M08 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: No helicity TMD matching channel is part of the accepted Collins ep assembly. A helicity boundary input is not a helicity TMD derivation. M08 source/export bindings ; executable comparison .					
beam_absorption = $\frac{C_F \left(-6L_Q^2 - 18L_Q - 48 + \pi^2 \right)}{12}$ fragment_absorption = $\frac{C_F \left(-6L_Q^2 - 18L_Q - 48 + \pi^2 \right)}{12}$ absorbed_hard = 0 M09 CSS ver- UNVERIFIED: no new expression sus separate-hard mapped. matching 1505.05589v1 Eq. (67) source $\delta C_{q \leftarrow q'}^{(\text{SIDIS})}(z, \mu_b)(x, \mu_b) = \delta_{q'q} \left[\delta(1 - x) + \frac{\alpha_s}{\pi} (-2C_F \delta(1 - x)) \right],$ No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection. Old: UNVERIFIED; paper: UNVERIFIED Continuation; same explicit map.					
M09 cont.	Same new-route scope.	Same accepted comparison.	1505.05589v1 Eq. (68) source $\delta \hat{C}_{q' \leftarrow q}^{(\text{SIDIS})}(z, \mu_b) = \delta_{q'q} \left[\delta(1 - z) + \frac{\alpha_s}{\pi} \left(\hat{P}_{q \leftarrow q}^c(z) \ln z - 2C_F \delta(1 - z) \right) \right],$	Continuation; same explicit map.	Same target qualifications.
M09 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: The actual finite hard absorption is explicit. Literal CSS coefficient arguments and full per-leg canonical mapping are not equated by this scalar identity alone; original source artifacts remain unchanged. M09 source/export bindings ; executable comparison .					

Evolution and declared external inputs

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
E03 Full Collins twist-three mixing	New expression in continuation.	Accepted expression in continuation.	1505.05589v1 Eq. (A6) source $\hat{P}_{q \rightarrow q}^c(\hat{\xi}) = C_F \left[\frac{2\hat{\xi}}{(1 - \hat{\xi})_+} + \frac{3}{2} \delta(1 - \hat{\xi}) + \dots \right],$	Only the actual two-fraction mixing contribution and its physical endpoint action are compared; this does not close full twist-three evolution. Generic contact coefficients remain unverified.	Old: EXACT; paper: UNVERIFIED
E03 new expression:					
delta = 0 D0 = 0 D1 = 0 $\text{regular} = -\frac{C_A u^2 v^2}{(v - 1)^2 (uv - 1)} - \frac{C_A u^2}{(v - 1)^2 (uv - 1)} + \frac{C_A uv}{(v - 1)^2 (uv - 1)} + \frac{C_A u}{(v - 1)^2 (uv - 1)} - \frac{2C_F u^3 v^3}{(v - 1)^2 (uv - 1)} + \frac{4C_F u^3 v^2}{(v - 1)^2 (uv - 1)} - \frac{2C_F u^3 v}{(v - 1)^2 (uv - 1)} - \frac{2C_F u^2 v}{(v - 1)^2 (uv - 1)} + \frac{2C_F u^2}{(v - 1)^2 (uv - 1)} + \frac{2C_F uv}{(v - 1)^2 (uv - 1)} - \frac{2C_F u}{(v - 1)^2 (uv - 1)}$					
E03 accepted expression (archived):					
$\text{regular} = -\frac{u \left(C_A uv^2 + C_A u - C_A v - C_A + 2C_F u^2 v^3 - 4C_F u^2 v^2 + 2C_F u^2 v + 2C_F uv - 2C_F u - 2C_F v + 2C_F \right)}{(v - 1)^2 (uv - 1)}$ delta = 0 D0 = 0 D1 = 0					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
E03 cont.	Same new-route scope.	Same accepted comparison.	Literal equation in continuation: 1012.3419 Eq. (30), source label evlH	Continuation; same explicit map.	Same target qualifications.
1012.3419 Eq. (30), source label evlH source $\frac{\partial \hat{H}(z_h, \mu^2)}{\partial \ln \mu^2} = \frac{\alpha_s}{2\pi} \int \frac{dz}{z} \left[A(\hat{z}) \hat{H}(z, \mu^2) + \int \frac{dz_1}{z_1^2} \text{PV} \left(\frac{1}{\frac{1}{z} - \frac{1}{z_1}} \right) B(z_h, z, z_1) \hat{H}_F(z, z_1, \mu^2) \right],$					
E03 provenance. Fresh connected D-dimensional HF numerator, physical measurement derivatives, actual Kira and native sphere, with separate radial/collinear/endpoint action. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Displayed here are the four bulk-u components of fresh two-fraction HF mixing. All twelve bulk/contact transport components are retained; the v-delta and v-derivative distributions appear in C15/C17. This is not a closed evolution system for every twist-three operator. C16 compares the color-resolved bulk to the two literal alternatives; C15 checks finite subtraction and scale consistency. New-minus-old: 0. Complete literature difference is unverified. E03 source/export bindings ; executable comparison .					
E04	One-loop UNVERIFIED: no new expression mapped. Sudakov A1 and B1; A2 external	$A1 = C_F$ $B1 = -\frac{3C_F}{2}$	2007.07281v1 Eq. (A13) source $A^{(1)} = C_F,$ $A^{(2)} = \frac{C_F}{2} \left[C_A \left(\frac{67}{18} - \frac{\pi^2}{6} \right) - \frac{10}{9} T_R n_f \right],$ $B^{(1)} = -\frac{3}{2} C_F.$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
E04 cont.	Same new-route scope.	Same accepted comparison.	1505.05589v1 Eq. (27) source $A^{(1)} = C_F, A^{(2)} = \frac{C_F}{2} \left[C_A \left(\frac{67}{18} - \frac{\pi^2}{6} \right) - \frac{10}{9} T_R n_f \right], B^{(1)} = -\frac{3}{2} C_F.$	Continuation; same explicit map.	Same target qualifications.
E04 provenance. Not established by the current new-route comparison record. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: A1/B1 inferred from freshly generated one-loop anomalous dimensions. Standard A2 and beta-function coefficients are external resummation inputs, not a fresh two-loop derivation. Archived v0.4.0 note: Only the displayed one-loop pair is checked symbolically here; A2 remains explicitly external in the source/input ledger. E04 source/export bindings ; executable comparison .					
E05	Per-TMD UNVERIFIED: no new expression mapped. Sudakov / NLL	$C_F \ell_Q^2 - 3C_F \ell_Q$	2007.07281v1 Eq. (A12) source $S_{\text{pert}}(Q, b) = \int_{\mu_b^2}^{Q^2} \frac{d\bar{\mu}^2}{\bar{\mu}^2} \left[A(\alpha_s(\bar{\mu})) \ln \frac{Q^2}{\bar{\mu}^2} + B(\alpha_s(\bar{\mu})) \right].$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
E05 cont.	Same new-route scope.	Same accepted comparison.	1505.05589v1 Eq. (26) source $S_{\text{pert}}(Q, b) = \int_{\mu_b^2}^{Q^2} \frac{d\bar{\mu}^2}{\bar{\mu}^2} \left[A(\alpha_s(\bar{\mu})) \ln \frac{Q^2}{\bar{\mu}^2} + B(\alpha_s(\bar{\mu})) \right],$	Continuation; same explicit map.	Same target qualifications.
E05 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Fresh Spert/a is displayed, with transport LQ here renamed ell_Q=ln(Q ² /mu _b ²), distinct from the hard-function logarithm. Each TMD carries exp(-Spert/2). The full running-coupling higher-order exponent is an external evolution organization; no historical numerical RG routine is rerun or promoted to exact symbolic equality. E05 source/export bindings ; executable comparison .					
E06	Literal JET UNVERIFIED: no new expression mapped. Sudakov measure	$C_F \ell_Q^2 - 3C_F \ell_Q$	1707.00913v2 Eq. (14) source $S_{\text{pert}}(Q, b) = \int_{\mu_b}^Q \frac{d\mu'}{\mu'} \left[A \ln \left(\frac{Q^2}{\mu'^2} \right) + B \right],$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
E06 provenance. Not established by the current new-route comparison record. Historical group status: MISMATCH. Historical conversion: Compare Spert/a at strict first order with fresh A/a=2CF and B/a=-3CF. ell_Q=ln(Q ² /mu _b ²). The literal dmu/mu is half dmu ² /mu ² ; integration gives CF ell_Q ² /2-3CF ell_Q/2. Archived v0.4.0 note: Literal measure discrepancy retained; no source correction is silently made. Our displayed polynomial is the actual fresh d14 fixed-coupling coefficient; no higher-order derivation is claimed. E06 source/export bindings ; executable comparison .					
E07	b-star and natural scale UNVERIFIED: no new expression mapped.	No fresh analytical expression exported	2007.07281v1 Eq. (A14) source $b_T \Rightarrow b_* = \frac{b_T}{\sqrt{1 + b_T^2/b_{\text{max}}^2}},$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
E07 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Historical model/prescription row retained. Fitted providers and their b-star/nonperturbative boundary routines are not evaluated or imported into this fresh symbolic campaign. The all-b TMD matrix elements remain symbolic; no fresh analytical equality of this unused model is claimed. E07 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
E08 Nonperturbative f1	UNVERIFIED: no new expression mapped.	No fresh analytical expression exported	1505.05589v1 Eq. (146) source $S_{\text{NP}}^{f_1}(Q, b) = S_{\text{NP}}^{h_1}(Q, b) = \frac{g_2}{2} \ln\left(\frac{b}{b_*}\right) \ln\left(\frac{Q}{Q_0}\right) + g_q b^2,$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
E08 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Historical model/prescription row retained. Fitted providers and their b-star/nonperturbative boundary routines are not evaluated or imported into this fresh symbolic campaign. The all-b TMD matrix elements remain symbolic; no fresh analytical equality of this unused model is claimed. E08 source/export bindings ; executable comparison .					
E09 Nonperturbative D1	UNVERIFIED: no new expression mapped.	No fresh analytical expression exported	1505.05589v1 Eq. (147) source $S_{\text{NP}}^{D_1}(Q, b) = \frac{g_2}{2} \ln\left(\frac{b}{b_*}\right) \ln\left(\frac{Q}{Q_0}\right) + \frac{g_h}{z^2} b^2,$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
E09 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Historical model/prescription row retained. Fitted providers and their b-star/nonperturbative boundary routines are not evaluated or imported into this fresh symbolic campaign. The all-b TMD matrix elements remain symbolic; no fresh analytical equality of this unused model is claimed. E09 source/export bindings ; executable comparison .					
E10 Nonperturbative collins	UNVERIFIED: no new expression mapped.	No fresh analytical expression exported	1505.05589v1 Eq. (148) source $S_{\text{NP}}^{\text{collins}}(Q, b) = \frac{g_2}{2} \ln\left(\frac{b}{b_*}\right) \ln\left(\frac{Q}{Q_0}\right) + \frac{g_h - g_c}{z^2} b^2,$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
E10 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Historical model/prescription row retained. Fitted providers and their b-star/nonperturbative boundary routines are not evaluated or imported into this fresh symbolic campaign. The all-b TMD matrix elements remain symbolic; no fresh analytical equality of this unused model is claimed. E10 source/export bindings ; executable comparison .					
E11 Transversity fitted boundary	UNVERIFIED: no new expression mapped.	No fresh analytical expression exported	1505.05589v1 Eq. (110) source $h_1^q(x, Q_0) = N_q^h x^{a_q} (1-x)^{b_q} \frac{(a_q + b_q)^{a_q + b_q}}{a_q^{a_q} b_q^{b_q}} \frac{1}{2} (f_1^q(x, Q_0) + g_1^q(x, Q_0)) ,$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
E11 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Historical model/prescription row retained. Fitted providers and their b-star/nonperturbative boundary routines are not evaluated or imported into this fresh symbolic campaign. The all-b TMD matrix elements remain symbolic; no fresh analytical equality of this unused model is claimed. E11 source/export bindings ; executable comparison .					
E12 Collins fitted boundary	UNVERIFIED: no new expression mapped.	No fresh analytical expression exported	1505.05589v1 Eq. (111) source $\hat{H}_{fav}^{(3)}(z, Q_0) = N_u^c z^{\alpha_u} (1-z)^{\beta_u} D_{\pi^+/u}(z, Q_0) ,$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
E12 cont.	Same new-route scope.	Same accepted comparison.	1505.05589v1 Eq. (112) source $\hat{H}_{unf}^{(3)}(z, Q_0) = N_d^c z^{\alpha_d} (1-z)^{\beta_d} D_{\pi^+/d}(z, Q_0) ,$	Continuation; same explicit map.	Same target qualifications.
E12 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Historical model/prescription row retained. Fitted providers and their b-star/nonperturbative boundary routines are not evaluated or imported into this fresh symbolic campaign. The all-b TMD matrix elements remain symbolic; no fresh analytical equality of this unused model is claimed. E12 source/export bindings ; executable comparison .					

Fourier conventions

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
F01 Unpolarized FF Fourier kernel	$\frac{1}{2\pi z^2}$	$\frac{1}{2\pi z^2}$	Literal equation in continuation: 1505.05589v1 Eq. (142)	Restore the independent azimuthal inverse-transform factor $1/(2\pi)$; the native rank-zero Jacobian is z^{-2} . The displayed scalar omits b db and J_0 .	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
1505.05589v1 Eq. (142) source $D_{h/q}(z, p_\perp^2; Q) = \frac{1}{z^2} \int_0^\infty \frac{db}{(2\pi)} \frac{b}{b} J_0(p_\perp b/z) \hat{C}_{i \leftarrow q}^{D_1} \otimes D_{h/i}(z, \mu_b) e^{-\frac{1}{2} S_{\text{pert}}(Q, b_*) - S_{\text{NP}}^{D_1}(Q, b)},$					
F01 provenance.	Actual native determinant of $p_\perp$ - z $k_\perp$; Fourier measure is a declared operator definition, not a generated QCD interaction. Historical group status: UNVERIFIED. Historical conversion: Remove b db J_0 from the radial transform; Fourier phase is $j \cdot b/z$ with $d^2b/(2\pi)^2$. New-minus-old: 0. New-minus-converted-reference: 0. Shared equation IDs: F01, F02. F01 source/export bindings ; executable comparison .				
F03 Collins rank-one kernel, sign, z and mass	radial = $\frac{1}{4\pi z^3}$ vector_without_i_b = $-\frac{1}{2z}$	radial = $\frac{1}{4\pi z^3}$ vector_without_i_b = $-\frac{1}{2z}$	Literal equation in continuation: 1505.05589v1 Eq. (143)	The native rank-one coefficient uses the physical Collins moment and $p_\perp$ - z $k_\perp$. Restore $1/(2\pi)$ radially; the Cartesian coefficient without i b follows the explicit Fourier derivative, $-z^2$ times the native rank-one ratio.	Old: EXACT; paper: UNVERIFIED
1505.05589v1 Eq. (143) source $\frac{p_\perp}{M_h} H_{1h/q}^\perp(z, p_\perp^2; Q) = \frac{1}{z^2} \int_0^\infty \frac{db}{(4\pi z)} \frac{b^2}{b^2} J_1(p_\perp b/z) \delta \hat{C}_{i \leftarrow q}^{\text{collins}} \otimes \hat{H}_{h/i}^{(3)}(z, \mu_b) e^{-\frac{1}{2} S_{\text{pert}}(Q, b_*) - S_{\text{NP}}^{\text{collins}}(Q, b)},$					
F03 cont.	Same new-route scope.	Same accepted comparison.	Literal equation in continuation: 1505.05589v1 Eq. (144)	Continuation; same explicit map.	Same target qualifications.
1505.05589v1 Eq. (144) source $\frac{p_\perp}{M_h} H_{1h/q}^\perp(z, p_\perp^2; Q) = -\frac{1}{z^2} \int_0^\infty \frac{db}{(2\pi)} \frac{b^2}{b^2} J_1(p_\perp b/z) \delta \hat{C}_{i \leftarrow q}^{\text{collins}} \otimes \hat{H}_{1h/j}^{\perp(1)}(z) _{\text{Trento}}(z, \mu_b) e^{-\frac{1}{2} S_{\text{pert}}(Q, b_*) - S_{\text{NP}}^{\text{collins}}(Q, b)},$					
F03 cont.	Same new-route scope.	Same accepted comparison.	Literal equation in continuation: 1505.05589v1 Eq. (145)	Continuation; same explicit map.	Same target qualifications.
1505.05589v1 Eq. (145) source $\frac{p_\perp}{zM_h} H_{1h/q}^\perp(z, p_\perp^2; Q) _{\text{Trento}} = \frac{1}{z^2} \int_0^\infty \frac{db}{(2\pi)} \frac{b^2}{b^2} J_1(p_\perp b/z) \delta \hat{C}_{i \leftarrow q}^{\text{collins}} \otimes \hat{H}_{1h/j}^{\perp(1)}(z) _{\text{Trento}}(z, \mu_b) e^{-\frac{1}{2} S_{\text{pert}}(Q, b_*) - S_{\text{NP}}^{\text{collins}}(Q, b)}.$					
F03 provenance.	Fresh physical Clifford moment and native rank-one Fourier/Jacobian algebra. Hadron-mass convention is part of the separately stated moment definition. Historical group status: UNVERIFIED. Historical conversion: The bound native vector and radial coefficients retain sign, mass and z factors. The signed angular transform and the native-to-Trento map are separate relations, not a factor chosen from a numerical plot. Archived v0.4.0 note: The independent N02/C16 operator review supports the native convention; full equality with every differently named paper Collins function remains qualified by C04. New-minus-old: 0. Complete literature difference is unverified. F03 source/export bindings ; executable comparison .				
F04 Hadron-in-jet dimensionful first moment	UNVERIFIED: no new expression mapped.	$-2Mhz$	Literal equation in continuation: 1707.00913v2 Eq. (6b)	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
1707.00913v2 Eq. (6b) source $\frac{j_\perp}{z_h M_h} H_{1h/q}^\perp(z_h, j_\perp^2; Q) = \frac{1}{z_h^2} \int_0^\infty \frac{db}{(2\pi)} \frac{b^2}{b^2} J_1(j_\perp b/z_h) \delta \hat{C}_{i \leftarrow q}^{\text{collins}} \otimes \hat{H}_{1h/i}^{\perp(1)}(z_h, \mu_b) e^{-\frac{1}{2} S_{\text{pert}}(Q, b_*) - S_{\text{NP}}^{\text{collins}}(Q, b)}.$					
F04 cont.	Same new-route scope.	Same accepted comparison.	Literal equation in continuation: 1505.05589v1 Eq. (144)	Continuation; same explicit map.	Same target qualifications.
1505.05589v1 Eq. (144) source $\frac{p_\perp}{M_h} H_{1h/q}^\perp(z, p_\perp^2; Q) = -\frac{1}{z^2} \int_0^\infty \frac{db}{(2\pi)} \frac{b^2}{b^2} J_1(p_\perp b/z) \delta \hat{C}_{i \leftarrow q}^{\text{collins}} \otimes \hat{H}_{1h/j}^{\perp(1)}(z) _{\text{Trento}}(z, \mu_b) e^{-\frac{1}{2} S_{\text{pert}}(Q, b_*) - S_{\text{NP}}^{\text{collins}}(Q, b)},$					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
F04 cont.	Same new-route scope.	Same accepted comparison.	Literal equation in continuation: 1505.05589v1 Eq. (145)	Continuation; same explicit map.	Same target qualifications.
1505.05589v1 Eq. (145) source $\frac{p_{\perp}}{zM_h} H_{1h/q}^{\perp}(z, p_{\perp}^2; Q) _{\text{Trento}} = \frac{1}{z^2} \int_0^{\infty} \frac{db}{(2\pi)} \frac{b^2}{(2\pi)} J_1(p_{\perp} b/z) \delta \hat{C}_{i \leftarrow q}^{\text{collins}} \otimes \hat{H}_{1h/j}^{\perp(1)}(z) _{\text{Trento}}(z, \mu_b) e^{-\frac{1}{2} S_{\text{pert}}(Q, b_*) - S_{\text{NP}}^{\text{collins}}(Q, b)}.$					
F04 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: Compare the actual d14 dimensionful moment relation; native Hhat3 is not silently identified with a differently normalized first moment. Archived v0.4.0 note: Native/Kang/KPSY operator conversion is derived in C01/C02 and retains the printed-placement qualification C04. F04 source/export bindings ; executable comparison .					
F05 Recoil in-verse measure	UNVERIFIED: no new expression mapped.	$\frac{1}{2\pi}$	Literal equation in continuation: 2007.07281v1 Eq. (17)	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
2007.07281v1 Eq. (17) source $F_{UU}^h = \sigma_0 H_q(Q, \mu) \sum_q e_q^2 \mathcal{G}_q^h(z_h, \vec{j}_T, p_T^{\text{jet}} R, \mu)$ $\times \int \frac{d^2 \vec{b}_T}{(2\pi)^2} e^{i \vec{q}_T \cdot \vec{b}_T} f_q^{\text{TMD}}(x, \vec{b}_T, \mu) S_q(\vec{b}_T, y_{\text{jet}}, R, \mu).$					
F05 provenance. Not established by the current new-route comparison record. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Internal Fourier-Jacobian relation z^2 times the fresh rank-zero fragmentation radial prefactor. The actual two radial measures are shown separately in the fresh full all-b operator. Archived v0.4.0 note: This is an algebraic measure relation checked against the bound full all-b operator, not a historical Gaussian fixture. F05 source/export bindings ; executable comparison .					
F06 Independent recoil and fragmentation b	UNVERIFIED: no new expression mapped.	Accepted expression in continuation.	Literal equation in continuation: 2007.07281v1 Eq. (17)	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
F06 accepted expression (archived): $\text{HoldComplete} \left[\exp(i q \text{RecoilVector} \cdot b \text{RecoilVector}) \exp \left(\frac{i j \text{Vector} \cdot b \text{FragmentVector}}{z h} \right) \right]$					
2007.07281v1 Eq. (17) source $F_{UU}^h = \sigma_0 H_q(Q, \mu) \sum_q e_q^2 \mathcal{G}_q^h(z_h, \vec{j}_T, p_T^{\text{jet}} R, \mu)$ $\times \int \frac{d^2 \vec{b}_T}{(2\pi)^2} e^{i \vec{q}_T \cdot \vec{b}_T} f_q^{\text{TMD}}(x, \vec{b}_T, \mu) S_q(\vec{b}_T, y_{\text{jet}}, R, \mu).$					
F06 cont.	Same new-route scope.	Same accepted comparison.	2007.07281v1 Eq. (18) source $\mathcal{G}_q^h(z_h, \vec{j}_T, p_T^{\text{jet}} R) = \int \frac{d^2 \vec{b}'_T}{(2\pi)^2} e^{i \vec{j}_T \cdot \vec{b}'_T / z_h} D_{h/q}^{\text{TMD}}(z_h, \vec{b}'_T, p_T^{\text{jet}} R).$	Continuation; same explicit map.	Same target qualifications.
F06 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: Fresh d16 exports these independent defining phases and d19 has two physical radial integrals. Complete literal equation equality remains qualified by fragment normalization and measured-variable identification. F06 source/export bindings ; executable comparison .					
F08 Executed in-dependent Gaussian Fourier fixture	UNVERIFIED: no new expression mapped.	No fresh analytical expression exported	1505.05589v1 Eq. (140) source $f_1^q(x, k_{\perp}^2; Q) = \int_0^{\infty} \frac{db}{(2\pi)} \frac{b}{(2\pi)} J_0(k_{\perp} b) C_{q \leftarrow i}^{f_1} \otimes f_1^i(x, \mu_b) e^{-\frac{1}{2} S_{\text{pert}}(Q, b_*) - S_{\text{NP}}^{f_1}(Q, b)},$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
F08 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: The Gaussian fixture has numerical evidence in the supporting report. No fresh analytical Gaussian-fixture export is supplied, so symbolic comparison remains pending; the quantity is not dropped from the inventory. F08 source/export bindings ; executable comparison .					

Symbolic UU/UT assembly

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
A01 UU factor-ized polynomial / through a	New expression in continuation.	Accepted expression in continuation.	Literal equation in continuation: 2007.07281v1 Eq. (17)	Strict first-order a=alpha_s/(2pi) product of the same native hard, measured factors and jet replacement. f1/h1/D1/C1 are bound in the native observable to the full-scale distributions; HF is added only in local small-b OPE. This scalar polynomial is not the full published operator identity.	Old: EXACT; paper: UNVERIFIED
A01 new expression:					
$UU0 = D_0 HUU f_0$ $UU1 = 2B_q C_F D_0 HUU Y f_0 + B_q C_F D_0 HUU f_0 r - C_F D_0 HUU L^2 f_0 - 3C_F D_0 HUU L f_0 - \frac{C_F D_0 HUU f_0 r^2}{2} - 8C_F D_0 HUU f_0 + \frac{\pi^2 C_F D_0 HUU f_0}{6} + D_0 HUU f_1 + D_1 HUU f_0$					
A01 accepted expression (archived):					
$UU0 = D_0 HUU f_0$ $UU1 = 2B_q C_F D_0 HUU Y f_0 + B_q C_F D_0 HUU f_0 r - C_F D_0 HUU L^2 f_0 - 3C_F D_0 HUU L f_0 - \frac{C_F D_0 HUU f_0 r^2}{2} - 8C_F D_0 HUU f_0 + \frac{\pi^2 C_F D_0 HUU f_0}{6} + D_0 HUU f_1 + D_1 HUU f_0$					
2007.07281v1 Eq. (17) source $F_{UU}^h = \sigma_0 H_q(Q, \mu) \sum_q e_q^2 \mathcal{G}_q^h(z_h, \vec{j}_T, p_T^{\text{jet}} R, \mu)$ $\times \int \frac{d^2 \vec{b}_T}{(2\pi)^2} e^{i\vec{q}_T \cdot \vec{b}_T} f_q^{\text{TMD}}(x, \vec{b}_T, \mu) S_q(\vec{b}_T, y_{\text{jet}}, R, \mu).$					
A01 provenance. Fresh r07 assembly contracts r03 and r06 native nodes with fixed recipes and separately stores physical Fourier/flavor convolution integrands. Historical group status: UNVERIFIED. Historical conversion: Strict first-order multiplication of actual hard, soft, intrinsic matching and additional jet packets. UT also has the separate HF convolution retained in the native observable. Archived v0.4.0 note: This is the exact internal first-order factorization polynomial. Complete published-equation identification remains UNVERIFIED because measured-variable, Fourier and signed operator identifications remain qualified; cited equations give the leading-power structure, not a published complete finite HF polynomial. New-minus-old: 0. Complete literature difference is unverified. A01 source/export bindings ; executable comparison .					
A02 UT factor-ized polynomial / through a	New expression in continuation.	Accepted expression in continuation.	2007.07281v1 Eq. (19) source $f_q(x, k_T) \rightarrow h_1^q(x, k_T),$	Strict first-order a=alpha_s/(2pi) product of the same native hard, measured factors and jet replacement. f1/h1/D1/C1 are bound in the native observable to the full-scale distributions; HF is added only in local small-b OPE. This scalar polynomial is not the full published operator identity.	Old: EXACT; paper: UNVERIFIED
A02 new expression:					
$UT0 = C_0 HUT h_0$ $UT1 = 2B_q C_0 C_F HUT Y h_0 + B_q C_0 C_F HUT h_0 r - C_0 C_F HUT L^2 h_0 - 3C_0 C_F HUT L h_0 - \frac{C_0 C_F HUT h_0 r^2}{2} - 8C_0 C_F HUT h_0 + \frac{\pi^2 C_0 C_F HUT h_0}{6} + C_0 HUT h_1 + C_1 HUT h_0$					
A02 accepted expression (archived):					
$UT0 = C_0 HUT h_0$ $UT1 = 2B_q C_0 C_F HUT Y h_0 + B_q C_0 C_F HUT h_0 r - C_0 C_F HUT L^2 h_0 - 3C_0 C_F HUT L h_0 - \frac{C_0 C_F HUT h_0 r^2}{2} - 8C_0 C_F HUT h_0 + \frac{\pi^2 C_0 C_F HUT h_0}{6} + C_0 HUT h_1 + C_1 HUT h_0$					
A02 cont.	Same new-route scope.	Same accepted comparison.	2007.07281v1 Eq. (20) source $D_{h/q}(z_h, j_T) \rightarrow \frac{j_T}{z_h M_h} H_{1h/q}^\perp(z_h, j_T),$	Continuation; same explicit map.	Same target qualifications.
A02 provenance. Fresh r07 assembly contracts r03 and r06 native nodes with fixed recipes and separately stores physical Fourier/flavor convolution integrands. Historical group status: UNVERIFIED. Historical conversion: Strict first-order multiplication of actual hard, soft, intrinsic matching and additional jet packets. UT also has the separate HF convolution retained in the native observable. Archived v0.4.0 note: This is the exact internal first-order factorization polynomial. Complete published-equation identification remains UNVERIFIED because measured-variable, Fourier and signed operator identifications remain qualified; cited equations give the leading-power structure, not a published complete finite HF polynomial. New-minus-old: 0. Complete literature difference is unverified. A02 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
A03 Collins asymmetry definition	$\frac{N_0}{U_0}$	$\frac{N_0}{U_0}$	2007.07281v1 Eq. (16) source $A_{UT}^{\sin(\phi_s-\phi_h)} = \frac{F_{UT}^{\sin(\phi_s-\phi_h)}}{F_{UU}^h}.$	U0 is nonzero. Form the ratio only after common linear flavor and Fourier integration; this is the coefficient of a^0 in a strict first-order expansion.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
A03 provenance. Native symbolic r07 ratio expansion, bound to the newly assembled numerator and denominator definitions. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: N0 and U0 denote the same flavor/Fourier/bin linear functional applied to the UT and UU leading terms. New-minus-old: 0. New-minus-converted-reference: 0. A03 source/export bindings ; executable comparison .					
A04 Strict expansion of asymmetry / a^2 remainder	$-\frac{N_0 U_1}{U_0^2} + \frac{N_1}{U_0}$	$\frac{-N_0 U_1 + N_1 U_0}{U_0^2}$	2007.07281v1 Eq. (16) source $A_{UT}^{\sin(\phi_s-\phi_h)} = \frac{F_{UT}^{\sin(\phi_s-\phi_h)}}{F_{UU}^h}.$	U0 is nonzero. Form the ratio only after common linear flavor and Fourier integration; this is the coefficient of a^1 in a strict first-order expansion.	Old: EXACT; paper: EXACT AFTER EXPLICIT CONVERSION
A04 provenance. Native symbolic r07 ratio expansion, bound to the newly assembled numerator and denominator definitions. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Taylor-expand (N0+a N1)/(U0+a U1) only through a; discard O(a^2). New-minus-old: 0. New-minus-converted-reference: 0. A04 source/export bindings ; executable comparison .					
A05 NGL and NLO/power terms	UNVERIFIED: no new expression complete mapped.	No fresh analytical expression exported	2007.07281v1 Eq. (12) source $F_{UU} = \sigma_0 H_q(Q, \mu) \sum_q e_q^2 J_q(p_T^{\text{jet}} R, \mu)$ $\times \int \frac{d^2 \vec{b}_T}{(2\pi)^2} e^{i\vec{q}_T \cdot \vec{b}_T} f_q^{\text{TMD}}(x, \vec{b}_T, \mu) S_q(\vec{b}_T, y_{\text{jet}}, R, \mu),$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
A05 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: NGL resummation beyond this order is external; unrestricted nonsingular fixed-order recoil, power corrections and a finite-R completion are absent. This is a scope boundary, not an additional agreement. A05 source/export bindings ; executable comparison .					

Fragmenting-jet conversion and radius scope

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
N12 Additional fragmenting-jet UU conversion	New expression in continuation.	Accepted expression in continuation.	1705.08443v2 Eq. (2.40), label eq:newD source $\mathcal{D}_{h/i}^R(z_h, \mathbf{b}; \mu) \equiv D_{h/i}(z_h, \mathbf{b}, \mu, \nu) S_i(\mathbf{b}, \mu, \nu R),$	The fragmenting-jet factor is compared with its measured TMD overlap at the declared jet rapidity scale Vstandard=V-g=0. This is the fragmenting-jet replacement, not the inclusive quark jet function.	Old: EXACT; paper: UNVERIFIED
N12 new expression: bare = $-\frac{B^2 C_F}{2} - \frac{2BC_F}{\eta} - \frac{\pi^2 C_F}{12} - \frac{2C_F}{\epsilon\eta} + \frac{C_F}{\epsilon^2}$ counterterm = 0 overlap = $-\frac{B^2 C_F}{2} - \frac{2BC_F}{\eta} - \frac{\pi^2 C_F}{12} - \frac{2C_F}{\epsilon\eta} + \frac{C_F}{\epsilon^2}$ finite = 0 independent_finite = 0 gamma_G_minus_TMD = 0 N12 accepted expression (archived): bare = $\frac{C_F \left(-\frac{\epsilon(6B\epsilon(B\eta+4)+\pi^2\epsilon\eta+24)}{\eta} + 12 \right)}{12\epsilon^2}$ counterterm = 0 overlap = $\frac{C_F \left(-\frac{\epsilon(6B\epsilon(B\eta+4)+\pi^2\epsilon\eta+24)}{\eta} + 12 \right)}{12\epsilon^2}$ finite = 0 independent_finite = 0 gamma_G_minus_TMD = 0					
N12 cont.	Same new-route scope.	Same accepted comparison.	1705.08443v2 Eq. (2.43), label eq:standard source $\hat{\mathcal{D}}_{h/i}(z_h, \mathbf{b}; \mu) \equiv D_{h/i}(z_h, \mathbf{b}, \mu, \nu) \sqrt{\hat{S}_i(\mathbf{b}, \mu, \nu)}.$	Continuation; same explicit map.	Same target qualifications.
N12 cont.	Same new-route scope.	Same accepted comparison.	Literal equation in continuation: 1705.08443v2 Eq. (2.33), label eq:softb	Continuation; same explicit map.	Same target qualifications.
1705.08443v2 Eq. (2.33), label eq:softb source $S_i(\mathbf{b}, \mu, \nu R) = \int d^2\boldsymbol{\lambda}_\perp e^{-i\boldsymbol{\lambda}_\perp \cdot \mathbf{b}} S_i(\boldsymbol{\lambda}_\perp, \mu, \nu R)$ $= 1 + \frac{\alpha_s}{2\pi} C_i \left[\frac{2}{\eta} \left(-\frac{1}{\epsilon} - \ln \left(\frac{\mu^2}{\mu_b^2} \right) \right) + \frac{1}{\epsilon^2} - \frac{1}{\epsilon} \ln \left(\frac{\nu^2 \tan^2(R/2)}{\mu^2} \right) \right.$ $\left. - \ln \left(\frac{\mu^2}{\mu_b^2} \right) \ln \left(\frac{\nu^2 \tan^2(R/2)}{\mu_b^2} \right) + \frac{1}{2} \ln^2 \left(\frac{\mu^2}{\mu_b^2} \right) - \frac{\pi^2}{12} \right].$					
N12 cont.	Same new-route scope.	Same accepted comparison.	1705.08443v2 Eq. (2.54), label eq:input source $\mathcal{D}_{h/i}^R(z_h, \mathbf{b}; \mu_b) = \hat{\mathcal{D}}_{h/i}(z_h, \mathbf{b}; \mu_b).$	Continuation; same explicit map.	Same target qualifications.
N12 provenance. Actual in-cone measured soft current and zero-bin overlap consume the same native scalar normalization. Bare terms remain visible; no additional inclusive J is multiplied into the physical assembly. Historical group status: EXACT. Historical conversion: After intrinsic TMD matching and common virtual/collinear/soft allocations, at zeta_J=pT_jet^2R^2; eta-before-epsilon and fixed positive zh. Archived v0.4.0 note: The additional finite remainder is zero in this organization. Ordinary NLO intrinsic TMD and soft/jet effects remain. No inclusive J multiplier or pp hard factor is imported. New-minus-old: 0. Complete literature difference is unverified. N12 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
N13 Additional fragmenting-jet UT conversion	New expression in continuation.	Accepted expression in continuation.	1705.08443v2 Eq. (2.40), label eq:newD source $\mathcal{D}_{h/i}^R(z_h, \mathbf{b}; \mu) \equiv D_{h/i}(z_h, \mathbf{b}, \mu, \nu) S_i(\mathbf{b}, \mu, \nu R),$	The fragmenting-jet factor is compared with its measured TMD overlap at the declared jet rapidity scale Vstandard=V-g=0. This is the fragmenting-jet replacement, not the inclusive quark jet function.	Old: EXACT; paper: UNVERIFIED
N13 new expression: bare = $-\frac{B^2 C_F}{2} - \frac{2BC_F}{\eta} - \frac{\pi^2 C_F}{12} - \frac{2C_F}{\epsilon\eta} + \frac{C_F}{\epsilon^2}$ counterterm = 0 overlap = $-\frac{B^2 C_F}{2} - \frac{2BC_F}{\eta} - \frac{\pi^2 C_F}{12} - \frac{2C_F}{\epsilon\eta} + \frac{C_F}{\epsilon^2}$ finite = 0 independent_finite = 0 gamma_G_minus_TMD = 0 N13 accepted expression (archived): bare = $\frac{C_F \left(-\frac{\epsilon(6B\epsilon(B\eta+4)+\pi^2\epsilon\eta+24)}{\eta} + 12 \right)}{12\epsilon^2}$ counterterm = 0 overlap = $\frac{C_F \left(-\frac{\epsilon(6B\epsilon(B\eta+4)+\pi^2\epsilon\eta+24)}{\eta} + 12 \right)}{12\epsilon^2}$ finite = 0 independent_finite = 0 gamma_G_minus_TMD = 0					
N13 cont.	Same new-route scope.	Same accepted comparison.	1705.08443v2 Eq. (2.43), label eq:standard source $\hat{\mathcal{D}}_{h/i}(z_h, \mathbf{b}; \mu) \equiv D_{h/i}(z_h, \mathbf{b}, \mu, \nu) \sqrt{\hat{S}_i(\mathbf{b}, \mu, \nu)}.$	Continuation; same explicit map.	Same target qualifications.
N13 cont.	Same new-route scope.	Same accepted comparison.	Literal equation in continuation: 1705.08443v2 Eq. (2.33), label eq:softb	Continuation; same explicit map.	Same target qualifications.
1705.08443v2 Eq. (2.33), label eq:softb source $S_i(\mathbf{b}, \mu, \nu R) = \int d^2\lambda_\perp e^{-i\lambda_\perp \cdot \mathbf{b}} S_i(\lambda_\perp, \mu, \nu R)$ $= 1 + \frac{\alpha_s}{2\pi} C_i \left[\frac{2}{\eta} \left(-\frac{1}{\epsilon} - \ln \left(\frac{\mu^2}{\mu_b^2} \right) \right) + \frac{1}{\epsilon^2} - \frac{1}{\epsilon} \ln \left(\frac{\nu^2 \tan^2(R/2)}{\mu^2} \right) - \ln \left(\frac{\mu^2}{\mu_b^2} \right) \ln \left(\frac{\nu^2 \tan^2(R/2)}{\mu_b^2} \right) + \frac{1}{2} \ln^2 \left(\frac{\mu^2}{\mu_b^2} \right) - \frac{\pi^2}{12} \right].$					
N13 cont.	Same new-route scope.	Same accepted comparison.	1705.08443v2 Eq. (2.54), label eq:input source $\mathcal{D}_{h/i}^R(z_h, \mathbf{b}; \mu_b) = \hat{\mathcal{D}}_{h/i}(z_h, \mathbf{b}; \mu_b).$	Continuation; same explicit map.	Same target qualifications.
N13 provenance. Actual in-cone measured soft current and zero-bin overlap consume the same native scalar normalization. Bare terms remain visible; no additional inclusive J is multiplied into the physical assembly. Historical group status: EXACT. Historical conversion: After intrinsic TMD matching and common virtual/collinear/soft allocations, at zeta_J=pT_jet^2R^2; eta-before-epsilon and fixed positive zh. Archived v0.4.0 note: The additional finite remainder is zero in this organization. Ordinary NLO intrinsic TMD and soft/jet effects remain. No inclusive J multiplier or pp hard factor is imported. New-minus-old: 0. Complete literature difference is unverified. N13 source/export bindings ; executable comparison .					
N14 Radius validity and fraction scope	UNVERIFIED: no new expression mapped.	No fresh analytical expression exported	2007.07281v1 Eq. (A2) source $J_q(p_T R, \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[\frac{1}{2} \ln^2 \left(\frac{\mu^2}{p_T^2 R^2} \right) + \frac{3}{2} \ln \left(\frac{\mu^2}{p_T^2 R^2} \right) + \frac{13}{2} - \frac{3}{4} \pi^2 \right],$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
N14 cont.	Same new-route scope.	Same accepted comparison.	2007.07281v1 Eq. (A3) source $S_q(\vec{b}_T, y_{\text{jet}}, R, \mu) = 1 + \frac{\alpha_s}{2\pi} C_F \left[-\ln\left(\frac{e^{-2y_{\text{jet}}}}{R^2}\right) \ln\left(\frac{\mu^2}{\mu_b^2}\right) - \frac{1}{2} \ln^2\left(\frac{1}{R^2}\right) \right],$	Continuation; same explicit map.	Same target qualifications.
N14 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: No change of convention is used for this comparison. Archived v0.4.0 note: The native derivation drops R^2 terms. Equality of the leading-power narrow-cone formula to printed A1–A3 does not certify R=1 or finite-fraction endpoints. N14 source/export bindings ; executable comparison .					

F-type operators, support and two-fraction matching

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
C01 Imaginary field 1 : 4HIMhz1 chiral-odd F projection / tree		$H_F^N(z, z_1) = 4 \operatorname{Im} \hat{H}_{FU} M_h z_1$	Literal equation in continuation: 1512.07233v2 Eq. (14)	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {}. No paper-equation equality is inferred from an old/new field match.	Old: UNVERIFIED; paper: UNVERIFIED
1512.07233v2 Eq. (14) source $\Delta_{F,ij}^{q,\rho}(z,z')=\frac{1}{N_c}\sum_X\int_{-\infty}^{\infty}\frac{d\lambda}{2\pi}\int_{-\infty}^{\infty}\frac{d\mu}{2\pi}\mathrm{e}^{i\frac{\lambda}{z'}+i(\frac{1}{z}-\frac{1}{z'})\mu}\times$ $\langle 0 \mathcal{W}[\pm\infty m;\mu m]igm_{\eta}F^{\eta\rho}(\mu m)\times$ $\mathcal{W}[\mu m;\lambda m]q_i(\lambda m) P_hS_h;X\rangle\times$ $\langle P_hS_h;X \bar{q}_j(0)\mathcal{W}[0;\pm\infty m] 0\rangle.$					
C01 cont.	field 2 : 2HIMhz1	$H_F^K(z, z_1) = 2 \operatorname{Im} \hat{H}_{FU} M_h z_1$	Literal equation in continuation: 1512.07233v2 Eq. (16)	Continuation; same explicit map.	Same target qualifications.
1512.07233v2 Eq. (16) source $\Delta_F^{q,\rho}(z,z')=$ $\frac{M_h}{z}\left(\epsilon^{P_hm\rho}S_h/P_h\,i\hat{D}_{FT}^q(z,z')\right.$ $\left.-(S_h^\rho-P_h^\rho(m\cdot S_h))/P_h\gamma_5\,\hat{G}_{FT}^q(z,z')\right.$ $\left.+\frac{i}{2}([/P_h,\gamma^\rho]-P_h^\rho[/P_h,/m])\,i\hat{H}_{FU}^q(z,z')\right.$ $\left.+\frac{M_h}{2}(m\cdot S_h)\,([/P_h,\gamma^\rho]\gamma_5-P_h^\rho[/P_h,/m]\gamma_5)\,\hat{H}_{FL}^q(z,z')\right).$					
C01 provenance. 2 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. Only a subset is matched; the whole group remains UNVERIFIED. Omitted scope: C01 third historical reference-projection dual trace is not separately exported in new helper. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: The external K15 Eq.14/16 correlator is projected with independently constructed physical gamma matrices; lambda=Ph+ xi, mu=Ph+ eta, m=n/Ph+. Field, transverse-index and imaginary-projection maps are explicit. New-minus-old: UNVERIFIED; no machine-readable old target bound. Complete literature difference is unverified. C01 source/export bindings ; executable comparison .					

C02 Native-to-Kang normalization and parent / tree	field 1 : 2	$\kappa_{\hat{H}} = 2$	0903.4680 Sec. 2 moment definition (unnumbered) source $\hat{H}(z) = \int d^2p_{\perp} \frac{p_{\perp}^2}{2M} H_1^{\perp}(z, p_{\perp})$	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {}. No paper-equation equality is inferred from an old/new field match.	Old: EXACT; paper: UNVERIFIED
C02 cont.	field 2 : 2	$\kappa_F = 2$	1012.3419 Eq. (6) source $\hat{H}(z) = \int d^2p_{\perp} \frac{ \vec{p}_{\perp} ^2}{M_h} H_1^{\perp}(z, p_{\perp}^2),$	Continuation; same explicit map.	Same target qualifications.

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
C02 cont.	field 3 : 1	$\kappa_{\hat{H}}/\kappa_F = 1$	1012.3419 Eq. (1), pinned source; spin-average and Wilson-link paragraph source $\Delta(z_h, p_\perp) = \frac{1}{z_h} \int \frac{d\xi^- d^2\xi_\perp}{(2\pi)^3} e^{ik^+\xi^- - i\vec{k}_\perp \cdot \vec{\xi}_\perp} \langle 0 \mathcal{L}_\xi \psi(\xi) PSX \rangle \langle PSX \bar{\psi}(0) \mathcal{L}_0^\dagger 0 \rangle,$	Continuation; same explicit map.	Same target qualifications.
C02 cont.	field 4 : $\frac{1}{8}\sqrt{kt}2u^3v$	$\mathcal{N}_{\text{parent,density}} = \frac{1}{8}\sqrt{k_T^2}u^3v$	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C02 provenance. 4 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: The same raw cut matrix is Delta_N=B/z and Delta_K=B/(2z). Both moment and F inverse density acquire a factor two; the matching ratio is one. The longitudinal Jacobian, parent moment and both actual conjugate placements are separately retained. New-minus-old: 0. Complete literature difference is unverified. C02 source/export bindings ; executable comparison .					
C03 Finite- endpoint Wilson transverse deriva- tive / through g2	New expression in continuation.	Accepted expression in continuation.	Literal equation in continuation: 0903.4680 Eq. (3)	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: Only bound ell/s Module serials are alpha-renamed to a common LOCAL serial in the held source definition; original snippets remain unchanged.; {}. No paper-equation equality is inferred from an old/new field match.	Old: EXACT AFTER EXPLICIT CONVERSION; paper: UNVERIFIED
C03 new expression: $(-gA_\alpha(L) + i\partial_\alpha)(W(L, \xi).\psi(\xi)) = g\text{Inactive[Integrate]}[W(L, s).F_{\alpha n}(s).W(s, \xi).\psi(\xi), \{s, \xi, L\}] + D_\alpha(\psi(\xi))W(L, \xi).i$					
C03 accepted expression (archived): $(-gA_\alpha(L) + i\partial_\alpha)(W(L, \xi).\psi(\xi)) = g\text{Inactive[Integrate]}[W(L, s).F_{\alpha n}(s).W(s, \xi).\psi(\xi), \{s, \xi, L\}] + D_\alpha(\psi(\xi))W(L, \xi).i$					
0903.4680 Eq. (3) source $\hat{H}(z) = n^+ z^2 \int \frac{d\xi^-}{2\pi} e^{ik^+\xi^-} \frac{1}{2} \left\{ \text{Tr} \sigma^{\alpha+} \langle 0 \left[iD_\perp^\alpha + \int_{\xi^-}^{+\infty} d\zeta^- gF^{\alpha+}(\zeta^-) \right] \psi(\xi) P_h X \rangle \right. \\ \left. \times \langle P_h X \bar{\psi}(0) 0 \rangle + h.c. \right\} ,$					
C03 cont.	New expression in continuation.	Accepted expression in continuation.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C03 new expression: $C_{H_F}^{(0)} = 0$					
C03 accepted expression (archived): $C_{H_F}^{(0)} = 0$					
C03 provenance. 2 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. For readable mathematical display, the structurally matched held identity uses the archived mathematical rendering with the explicitly checked common bound/function names. The unchanged new held InputForm and its exact structural comparison remain linked. Historical group status: EXACT. Historical conversion: Finite-endpoint Wilson derivative is an internal free-associative identity through g^2 plus a transport-ODE proof; YZ Hhat is contextual motivation, not a literal copy of this finite-endpoint equation. The tree independent HF remainder is zero only after the whole rank-one operator is identified as Hhat. New-minus-old: 0. Complete literature difference is unverified. C03 source/export bindings ; executable comparison .					
C04 Literal D/link placement and upper-index operator	New expression in continuation.	Accepted expression in continuation.	1012.3419 Eq. (1), pinned source; spin-average and Wilson-link paragraph source $\Delta(z_h, p_\perp) = \frac{1}{z_h} \int \frac{d\xi^- d^2\xi_\perp}{(2\pi)^3} e^{ik^+\xi^- - i\vec{k}_\perp \cdot \vec{\xi}_\perp} \langle 0 \mathcal{L}_\xi \psi(\xi) PSX \rangle \langle PSX \bar{\psi}(0) \mathcal{L}_0^\dagger 0 \rangle,$	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {}. No paper-equation equality is inferred from an old/new field match.	Old: EXACT; paper: UNVERIFIED
C04 new expression: $D_N = \partial + igA_N, \quad D_R = \partial - igA_R, \quad A_R = -A_N$					
C04 accepted expression (archived): $\{\text{DNative} = \text{partial} + i\text{AN}g, \text{DReference} = \text{partial} - i\text{AR}g, \text{AR} = -\text{AN}, \text{WNative}(b, a) = \text{PathOrderedExp}(-ig\text{LineIntegral}(\text{AN}, \{a, b\})), \text{WReference}(b, a) = \text{PathOrderedExp}(ig\text{LineIntegral}(\text{AR}, \{a, b\}))\}$					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
C04 cont.	New expression in continuation.	Same accepted comparison.	Literal equation in continuation: 0903.4680 Eq. (3)	Continuation; same explicit map.	Same target qualifications.
C04 new expression: $W_N(b, a) = \mathcal{P} \exp[-ig \int_a^b A_N], \quad W_R(b, a) = \mathcal{P} \exp[+ig \int_a^b A_R]$					
0903.4680 Eq. (3) source $\hat{H}(z) = n^+ z^2 \int \frac{d\xi^-}{2\pi} e^{ik^+ \xi^-} \frac{1}{2} \left\{ \text{Tr} \sigma^{\alpha+} \langle 0 \left[i D_{\perp}^{\alpha} + \int_{\xi^-}^{+\infty} d\zeta^- g F^{\alpha+}(\zeta^-) \right] \psi(\xi) P_h X \rangle \right. \\ \left. \times \langle P_h X \bar{\psi}(0) 0 \rangle + h.c. \right\} ,$					
C04 cont.	New expression in continuation.	Same accepted comparison.	0903.4680 Eq. (4) source $\hat{H}_D(z_1, z_2) = n^+ z_1 z_2 \int \frac{d\xi^- d\zeta^-}{(2\pi)^2} e^{ik_2^+ \xi^-} e^{ik_g^+ \zeta^-} \frac{1}{2} \left\{ \text{Tr} \sigma^{\alpha+} \langle 0 i D_{\perp}^{\alpha}(\zeta^-) \psi(\xi^-) P_h X \rangle \right. \\ \left. \times \langle P_h X \bar{\psi}(0) 0 \rangle + h.c. \right\} ,$	Continuation; same explicit map.	Same target qualifications.
C04 new expression: $\Delta_{\text{connection}} = \Delta_{\text{linear}} = \Delta_{\text{commutator}} = \Delta_{\text{transport}} = \Delta_{\text{index}} = 0$					
C04 provenance. 2 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. Display aliases N/R denote the native/reference symbols; P exp and the integral denote the recorded PathOrderedExp and LineIntegral definitions with the same a-to-b endpoints. This is a covariant definition map, not a repair of literal paper placements. Historical group status: UNVERIFIED. Historical conversion: A_K=-A_N implies F_K=-F_N and matches covariant transport exponents with the displayed transverse-index map. The printed negative-sign K11 link and its placement are not proved by that conversion. A dagger alone does not repair endpoints/placements. New-minus-old: 0. Complete literature difference is unverified. C04 source/export bindings ; executable comparison .					
C05 complete-state insertion	First field 1 : $m(k-w-y) + l(w-x+y-1)$	$\Theta_1 = m(k-w-y) + l(w-x+y-1)$	Literal equation in continuation: 0812.3783v2 Eq. (3)	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {}. No paper-equation equality is inferred from an old/new field match.	Old: EXACT; paper: UNVERIFIED
0812.3783v2 Eq. (3) source $\Delta_F^i(\frac{1}{z}, \frac{1}{z'}) = \sum_{X,Y} \int \frac{d\xi^+}{2\pi} \frac{d\eta^+}{2\pi} e^{i\frac{1}{z'} P^- \xi^+ + i(\frac{1}{z} - \frac{1}{z'}) P^- \eta^+} \\ \times \frac{1}{3} \text{Tr} \left[\langle 0 g t_a F_a^{-i}(\eta^+) Y \rangle \right. \\ \left. \times \langle Y \psi(\xi^+) P, X \rangle \langle P, X \bar{\psi}(0^+) 0 \rangle \right] \\ = \sum_{X,Y} \delta \left(\left(\frac{1}{z} - 1 \right) P^- - \sum_i p_i^- \right) \\ \times \delta \left(\left(\frac{1}{z} - \frac{1}{z'} \right) P^- - \sum_j q_j^- \right) \\ \times \frac{1}{3} \text{Tr} \left[\langle 0 g t_a F_a^{-i}(0^+) Y \rangle \right. \\ \left. \times \langle Y \psi(0^+) P, X \rangle \langle P, X \bar{\psi}(0^+) 0 \rangle \right] .$					
C05 cont.	field 2 : $\{ \{ x \rightarrow k-1, y \rightarrow k-w \} \}$	$\{ \{ x \rightarrow k-1, y \rightarrow k-w \} \}$	0812.3783v2 Eq. (4) source $\frac{1}{z} \geq 1 \quad \text{and} \quad \frac{1}{z'} \geq \frac{1}{z'} .$	Continuation; same explicit map.	Same target qualifications.
C05 cont.	field 3 : $k \geq 1 \wedge w \leq k$	$k \geq 1 \wedge w \leq k$	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C05 provenance. 3 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Native k=1/z and w=1/z1; l,m are the dimensionless light-front Fourier coordinates. Solving the fresh first phase gives x=k-1 and y=k-w, hence k>=1,w<=k. New-minus-old: 0. Complete literature difference is unverified. C05 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
C06 Second in- sertion and both pole exclusions	field 1 : $m(k - w - x + y - 1) + l(w - y)$	$\Theta_2 = m(k - w - x + y - 1) + l(w - y)$	Literal equation in continuation: 0812.3783v2 Eq. (8)	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {}. No paper-equation equality is inferred from an old/new field match.	Old: EXACT; paper: UNVERIFIED
0812.3783v2 Eq. (8) source	$\Delta_F^i(\frac{1}{z}, \frac{1}{z'}) = \sum_{X,Y} \delta\left(\left(\frac{1}{z} - 1\right)P^- - \sum_i p_i^-\right) \times \delta\left(\frac{1}{z'}P^- - \sum_j q_j^-\right) \times \frac{1}{3} \text{Tr}\left[\langle 0 t_a \psi(0^+) Y \rangle \times \langle Y g F_a^{-i}(0^+) P, X \rangle \langle P, X \bar{\psi}(0^+) 0 \rangle\right].$				
C06 cont.	field 2 : $\{\{x \rightarrow k - 1, y \rightarrow w\}\}$	$\{\{x \rightarrow k - 1, y \rightarrow w\}\}$	0812.3783v2 Eq. (9) source $\frac{1}{z} \geq 1 \quad \text{and} \quad \frac{1}{z'} > 0.$	Continuation; same explicit map.	Same target qualifications.
C06 cont.	field 3 : $\text{HoldComplete}[k \geq 1 \wedge 0 < w < k]$	$k \geq 1 \wedge 0 < w < k$	0812.3783v2 Eq. (10) source $0 \leq z \leq 1 \quad \text{and} \quad 0 < \frac{z}{z'} < 1.$	Continuation; same explicit map.	Same target qualifications.
C06 cont.	Same new-route scope.	Same accepted comparison.	0812.3783v2 Eq. (11) source $\Delta_F^i(\frac{1}{z}, \frac{1}{z}) = \Delta_F^i(\frac{1}{z}, 0) = 0,$	Continuation; same explicit map.	Same target qualifications.

C06 provenance. 3 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Second state insertion gives x=k-1,y=w. Combine both nonnegative supports and separately exclude the gluon and quark partonic poles. New-minus-old: 0. Complete literature difference is unverified. [C06 source/export bindings](#); [executable comparison](#).

C07 Gluon-pole tensor zero	New expression in continuation.	Accepted expression in continuation.	0812.3783v2 Eq. (5) source $M^{-i}(q_j) = \langle 0 g t_a F_a^{-i}(0^+) Y \rangle$	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {}. No paper-equation equality is inferred from an old/new field match.	Old: EXACT; paper: UNVERIFIED
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C07 new expression:

field 1 : $-\text{aaqmIqnMinus} + \text{aaqmMinusqnI} + \text{bbenIqmMinus} - \text{bbenMinusqmI} - \text{ccemIenMinus} + \text{ccemMinusenI}$

C07 accepted expression (archived):

$\mathcal{F}^{-i} = -\text{aaqmIqnMinus} + \text{aaqmMinusqnI} + \text{bbenIqmMinus} - \text{bbenMinusqmI} - \text{ccemIenMinus} + \text{ccemMinusenI}$

C07 cont.	New expression in continuation.	Accepted expression in continuation.	0812.3783v2 Eq. (6) source $M^{\mu\nu}(q_j) = \sum_{m,n} \left[q_m^\mu q_n^\nu A_{mn}(q_j) + q_m^\mu \epsilon^\nu(q_n) B_{mn}(q_j) + \epsilon^\mu(q_m) \epsilon^\nu(q_n) C_{mn}(q_j) - \{\mu \leftrightarrow \nu\} \right].$	Continuation; same explicit map.	Same target qualifications.
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C07 new expression:

field 2 : 0

C07 accepted expression (archived):

$\mathcal{F}^{-i} \big|_{q_m^- = q_n^- = e_m^- = e_n^- = 0} = 0$

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
C07 cont.	Same new-route scope.	Same accepted comparison.	0812.3783v2 Eq. (7) source $\frac{1}{z} \geq 1 \quad \text{and} \quad \frac{1}{z} > \frac{1}{z'}.$	Continuation; same explicit map.	Same target qualifications.
C07 provenance. 2 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: The fresh arbitrary-coefficient antisymmetric tensor vanishes when all constituent and physical-polarization minus components vanish. This is stronger than setting only the total momentum to zero. New-minus-old: 0. Complete literature difference is unverified. C07 source/export bindings ; executable comparison .					
C08 Inverse- fraction deriva- tive before the diagonal	field 1 : $i(l - m)$	$e^{-i[lw+(k-w)m]} \partial_w e^{i[lw+(k-w)m]} = i(l - m)$	Literal equation in continuation: 1512.07233v2 Eq. (A1)	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {}. No paper-equation equality is inferred from an old/new field match.	Old: EXACT; paper: UNVERIFIED
1512.07233v2 Eq. (A1) source $\Delta_{F,ij}^{q,\rho}(z, z') = \frac{1}{N_c} \sum_X \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int_{-\infty}^{\infty} \frac{d\mu}{2\pi} e^{i\frac{\lambda}{z'} + i(\frac{1}{z} - \frac{1}{z'})\mu}$ $\times \langle 0 ig m_\eta F^{\eta\rho}(\mu m) q_i(\lambda m) P_h S_h ; X \rangle \langle P_h S_h ; X \bar{q}_j(0) 0 \rangle.$					
C08 cont.	Same new-route scope.	Same accepted comparison.	Literal equation in continuation: 1512.07233v2 Eq. (A2)	Continuation; same explicit map.	Same target qualifications.
1512.07233v2 Eq. (A2) source $\left. \frac{\partial \Delta_{F,ij}^{q,\rho}(z, z')}{\partial(1/z')} \right _{z=z'} = \frac{1}{N_c} \sum_{X,Y} \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int_{-\infty}^{\infty} \frac{d\mu}{2\pi} i(\lambda - \mu)$ $\times e^{i\lambda(\frac{1}{z} - 1 + (P_Y - P_X) \cdot m)} e^{-i\mu P_Y \cdot m} \langle 0 ig m_\eta F^{\eta\rho}(0) Y \rangle$ $\times \langle Y q_i(0) P_h S_h ; X \rangle \langle P_h S_h ; X \bar{q}_j(0) 0 \rangle$ $= \frac{1}{N_c} \sum_{X,Y} \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} i\lambda e^{i\lambda(\frac{1}{z} - 1 - P_X \cdot m)} \delta(P_Y \cdot m) \langle 0 ig m_\eta F^{\eta\rho}(0) Y \rangle$ $\times \langle Y q_i(0) P_h S_h ; X \rangle \langle P_h S_h ; X \bar{q}_j(0) 0 \rangle$ $+ \frac{1}{N_c} \sum_{X,Y} \int_{-\infty}^{\infty} \frac{d\mu}{2\pi} (-i\mu) \delta(1/z - 1 + (P_Y - P_X) \cdot m) e^{-i\mu P_Y \cdot m}$ $\times \langle 0 ig m_\eta F^{\eta\rho}(0) Y \rangle \langle Y q_i(0) P_h S_h ; X \rangle \langle P_h S_h ; X \bar{q}_j(0) 0 \rangle.$					
C08 provenance. 1 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: Differentiation with respect to inverse fraction w inserts i(l-m), before the diagonal is imposed. The subsequent derivative theorem requires the full differentiated operator and spectral argument. New-minus-old: 0. Complete literature difference is unverified. C08 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
C09 cont.	New expression in continuation.	Accepted expression in continuation.	Literal equation in continuation: 1512.07233v2 Eq. (A4)	Continuation; same explicit map.	Same target qualifications.
C09 new expression: $J_{\text{last}}(y) = \frac{1}{2(y - \text{rest})}$					
C09 accepted expression (archived): $J_{\text{last}}(y) = \frac{1}{2(y - \text{rest})}$					
1512.07233v2 Eq. (A4) source					
$\begin{aligned} Boundary\ terms &= \frac{1}{N_c} \rlap{-}\!\!\!\!\!\int\limits_X \rlap{-}\!\!\!\!\!\int\limits_{Y_1} \cdots \rlap{-}\!\!\!\!\!\int\limits_{Y_{N-1}} \sum_Y \int \frac{d^2 \vec P_{YT}}{2P_Y \cdot m (2\pi)^3} \\ &\times \int_{-\infty}^\infty \frac{d\lambda}{2\pi} \int_{-\infty}^\infty \frac{d\mu}{2\pi} e^{i\lambda(1/z-1+(P_Y-P_X)\cdot m)} e^{-i\mu P_Y\cdot m} \\ &\times \langle 0 ig\,m_\eta F^{\eta\rho}(0) Y \rangle \langle Y q_i(0) P_h S_h ; X \rangle \\ &\times \left. \langle P_h S_h ; X \bar q_j(0) 0 \rangle \right _{P_Y\cdot m=(P_{Y_1}+\cdots+P_{Y_{N-1}})\cdot m} \end{aligned}$					
C09 provenance. 2 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. For readable mathematical display, the structurally matched held identity uses the archived mathematical rendering with the explicitly checked common bound/function names. The unchanged new held InputForm and its exact structural comparison remain linked. Historical group status: UNVERIFIED. Historical conversion: The full product rule retains the derivative of the state density, Jacobian and both boundaries. With y=sum_i y_i the last-particle measure is 1/[2(y-rest)]. K15 A3/A4 are source context; exact equality to its printed total-P_Y denominator is not certified. New-minus-old: 0. Complete literature difference is unverified. C09 source/export bindings ; executable comparison .					
C10 Actual na-tive two-fraction chain rule	field 1 :	$\frac{4\text{Mhff}(\xi,w)}{w}$	$H_F^N(\xi,1/w)=\frac{4M_hG(\xi,w)}{w}$	Literal equation in continuation: 1512.07233v2 discussion following Eq. (16)	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {ff -> G}. No paper-equation equality is inferred from an old/new field match. Old: EXACT AFTER EXPLICIT CONVERSION; paper: UNVERIFIED
1512.07233v2 discussion following Eq. (16) source Literal inline statements: $\Delta_F^\rho(z,z)=0$ $\Delta_F^\rho(z,0)=0$ $\frac{\partial}{\partial(1/z')}\Delta_F^\rho(z,z')\Big _{z'=z}=0$ $0\leq z\leq 1$ $z< z'<\infty$					
C10 cont.	field 2 :	$\frac{4\text{Mh}\xi\text{ff}\left(\xi,\frac{v}{\xi}\right)}{uv}$	$\Phi_u(v)=\frac{4M_h\xi G\left(\xi,\frac{v}{\xi}\right)}{uv}$	No additional published counterpart.	Continuation; same explicit map. Same target qualifications.
C10 cont.	field 3 :	$\frac{4\text{Mh}\xi\text{ff}\left(\xi,\frac{1}{\xi}\right)}{u}$	$\Phi_u(1)=\frac{4M_h\xi G\left(\xi,\frac{1}{\xi}\right)}{u}$	No additional published counterpart.	Continuation; same explicit map. Same target qualifications.
C10 cont.	field 4 :	$\frac{4\text{Mhff}^{(0,1)}\left(\xi,\frac{1}{\xi}\right)}{u}-\frac{4\text{Mh}\xi\text{ff}\left(\xi,\frac{1}{\xi}\right)}{u}$	$\Phi'_u(1)=\frac{4M_h\left(G^{(0,1)}\left(\xi,\frac{1}{\xi}\right)-\xi G\left(\xi,\frac{1}{\xi}\right)\right)}{u}$	No additional published counterpart.	Continuation; same explicit map. Same target qualifications.
C10 provenance. 4 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. Historical group status: EXACT AFTER EXPLICIT CONVERSION. Historical conversion: G=Im H_FU; xi=zh/u; actual F-to-F map is 4Mh G/w. Applying the two independent source endpoint identities gives both Phi(1)=0 and Phi-prime(1)=0. New-minus-old: 0. Complete literature difference is unverified. C10 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
C11 Generic contacts physical double-zero action	New expression in continuation.	Accepted expression in continuation.	1012.3419 Eq. (10) source $\hat{H}_D(z_1, z_2) = \text{PV} \left(\frac{1}{\frac{1}{z_1} - \frac{1}{z_2}} \right) \hat{H}_F(z_1, z_2) + \delta \left(\frac{1}{z_1} - \frac{1}{z_2} \right) \frac{z_1}{z_2} \hat{H}(z_2),$	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {ff -> G}; {}. No paper-equation equality is inferred from an old/new field match.	Old: EXACT AFTER EXPLICIT CONVERSION; paper: UNVERIFIED
C11 new expression: field 1 : $\frac{4c_0\text{Mh}\xi\text{ff}\left(\xi, \frac{1}{\xi}\right)}{u} - c_1 \left(\frac{4\text{Mhff}^{(0,1)}\left(\xi, \frac{1}{\xi}\right)}{u} - \frac{4\text{Mh}\xi\text{ff}\left(\xi, \frac{1}{\xi}\right)}{u} \right)$					
C11 accepted expression (archived): $\int_0^1 K[u, v] \left(-\text{Phi}(u, 1) + \text{Phi}(u, v) - (v - 1)\text{Phi}^{(0,1)}(u, 1) \right) dv + c_0(u)\text{Phi}(u, 1) - c_1(u)\text{Phi}^{(0,1)}(u, 1)$					
C11 cont.	New expression in continuation.	Accepted expression in continuation.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C11 new expression: field 2 : 0					
C11 accepted expression (archived): $c_0\Phi_u(1) - c_1\Phi'_u(1) = \frac{4c_0M_h\xi G\left(\xi, \frac{1}{\xi}\right)}{u} - \frac{4c_1M_h\left(G^{(0,1)}\left(\xi, \frac{1}{\xi}\right) - \xi G\left(\xi, \frac{1}{\xi}\right)\right)}{u}$					
C11 cont.	New expression in continuation.	Accepted expression in continuation.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C11 new expression: field 3 : $\text{HoldComplete} \left[\int_0^1 \left(-(v - 1)\text{Phi}^{(0,1)}(u, 1) + \text{Phi}(u, v) - \text{Phi}(u, 1) \right) K[u, v] dv + c_0(u)\text{Phi}(u, 1) - c_1(u)\text{Phi}^{(0,1)}(u, 1) \right]$					
C11 accepted expression (archived): $[c_0\Phi_u(1) - c_1\Phi'_u(1)]_{G=G_w=0} = 0$					
C11 provenance. 3 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. Historical group status: EXACT. Historical conversion: The full two-subtraction action and arbitrary c0/c1 precede physical reduction. The actual mapped contact action vanishes after both source endpoint identities. New-minus-old: 0. Complete literature difference is unverified. C11 source/export bindings ; executable comparison .					
C12 Damping prescriptions and sufficient endpoint class	New expression in continuation.	Accepted expression in continuation.	No additional published counterpart.	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {alphaH -> alpha}; {lambda -> lam}. No paper-equation equality is inferred from an old/new field match.	Old: EXACT AFTER EXPLICIT CONVERSION; paper: UNVERIFIED
C12 new expression: field 1 : $\left\{ -\frac{1}{\lambda^2 + 1} + \frac{1}{2} \log \left(\lambda^2 + 1 \right) - \log(\lambda), -\frac{1}{\lambda^2 + 1} - 2\lambda \cot^{-1}(\lambda) + 2 \right\}$					
C12 accepted expression (archived): $\{I_1^{++}, I_2^{++}\} = \left\{ -\frac{1}{\lambda^2 + 1} + \frac{1}{2} \log \left(\lambda^2 + 1 \right) - \log(\lambda), -\frac{1}{\lambda^2 + 1} - 2\lambda \cot^{-1}(\lambda) + 2 \right\}$					
C12 cont.	New expression in continuation.	Accepted expression in continuation.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C12 new expression: field 2 : $\left\{ \frac{1}{2} \log \left(\frac{1}{\lambda^2} + 1 \right), 1 - \lambda \tan^{-1} \left(\frac{1}{\lambda} \right) \right\}$					
C12 accepted expression (archived): $\{I_1^{+-}, I_2^{+-}\} = \left\{ \frac{1}{2} \log \left(\frac{1}{\lambda^2} + 1 \right), 1 - \lambda \cot^{-1}(\lambda) \right\}$					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
C12 cont.	New expression in continuation.	Accepted expression in continuation.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C12 new expression: field 3 : $\frac{1}{\text{alphaH}}$					
C12 accepted expression (archived): $\int_0^1 x^{\alpha-1} dx = \frac{1}{\alpha}$					
C12 provenance. 3 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. Historical group status: EXACT. Historical conversion: Independent exact rational integration reproduces both common-damping prescription moments. Linear tests retain a finite rescaling shift; quadratic tests have the same limit. A uniform Holder or Dini bound controls the limit. New-minus-old: 0. Complete literature difference is unverified. C12 source/export bindings ; executable comparison .					
C13 Regularity counterexamples and joint endpoint	field 1 : $\{0,0\}$	$\Phi(1-x) = x/\log(1/x), \quad (\Phi(1), \Phi'(1)) = \{0,0\}$	No additional published counterpart.	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {}. No paper-equation equality is inferred from an old/new field match.	Old: UNVERIFIED; paper: UNVERIFIED
C13 cont.	field 2 : $\log\left(\log\left(\frac{1}{\text{cut}}\right)\right)$	$\int_{\delta}^{\text{const}} \frac{dx}{x \log(1/x)} \sim \log(-\log(\delta))$	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C13 cont.	field 3 : $\frac{1}{\log\left(\frac{1}{w}\right)}$	$G(w) = \frac{1}{\log\left(\frac{1}{w}\right)}$	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C13 cont.	Same new-route scope.	$0 \leq \frac{(x^2 - 2x + 2)y + (1-x)x}{x(-y) + x + y} \leq 2$	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C13 provenance. 3 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. Only a subset is matched; the whole group remains UNVERIFIED. Omitted scope: C13 historical hardcoded-bulk joint diagnostic was intentionally excluded; the new actual-current endpoint decomposition is separate evidence. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. Historical group status: EXACT. Historical conversion: The derivative-flat x/log(1/x) test still gives a log-log divergence. G=1/log(1/w) similarly vanishes but G/w is nonintegrable. The displayed coupled-endpoint ratio is bounded though its directional limit is not unique. New-minus-old: UNVERIFIED; no machine-readable old target bound. Complete literature difference is unverified. C13 source/export bindings ; executable comparison .					
C14 Laurent extraction and strict NLO scheme counting	UNVERIFIED: no new expression	Accepted expression in continuation.	Literal equation in continuation: 1012.3419 Eq. (30), source label evlH	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
C14 accepted expression (archived): $\frac{x^{\epsilon+1}}{\epsilon} = \frac{x}{\epsilon} + x \log(x) + O(\epsilon)$					
1012.3419 Eq. (30), source label evlH source $\frac{\partial \hat{H}(z_h, \mu^2)}{\partial \ln \mu^2} = \frac{\alpha_s}{2\pi} \int \frac{dz}{z} \left[A(\hat{z}) \hat{H}(z, \mu^2) + \int \frac{dz_1}{z_1^2} \text{PV} \left(\frac{1}{\frac{1}{z} - \frac{1}{z_1}} \right) B(z_h, z, z_1) \hat{H}_F(z, z_1, \mu^2) \right],$					
C14 cont.	Same new-route scope.	Accepted expression in continuation.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C14 accepted expression (archived): $CZ^{-1} _{O(a)} = \{\text{aPT}(\text{cHat1} - \text{cHat0zHH}) + \text{cHat0}, \text{aPT}(\text{CHF1} - \text{cHat0zHF})\}$					
C14 cont.	Same new-route scope.	Accepted expression in continuation.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C14 accepted expression (archived): $\delta(C_{H_F} H_F) = a^2 \text{cHFonefiniteSchemeShift}$					
C14 provenance. Not established by the current new-route comparison record. Historical group status: EXACT. Historical conversion: The epsilon-dependent family is derivative-flat for epsilon>0 but its Laurent pole is linear. Finite O(a) HF scheme changes affect a*C_HF^(1) only at a^2 after actual pole cancellation; Hhat<-HF mixing remains at O(a). C14 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
C15 Connected D-dimensional HF bare, pole and finite coefficient	New expression in continuation.	$P_0 = -\frac{2u(uv-u+1)}{v-1}C_F + -\frac{u(uv^2+u-v-1)}{(v-1)^2(uv-1)}C_A$	0903.4680 Eq. (6) source $H_1^\perp(z_h, p_\perp) = \frac{\alpha_s}{2\pi^2} \frac{2M}{(p_\perp^2)^2} \int \frac{dz}{z} \left[A + \delta(\hat{\xi} - 1) \hat{H}(z) C_F \ln \frac{\hat{\xi}^2}{p_\perp^2} \right],$	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {}. No paper-equation equality is inferred from an old/new field match.	Old: EXACT; paper: UNVERIFIED
C15 new expression: $K_N = -\frac{C_A u(uv^2+u-v-1)}{(v-1)^2(uv-1)} - \frac{2C_F u(uv-u+1)}{v-1}$					
C15 cont.	New expression in continuation.	$P_1 = -2(u-1)C_F + -\frac{(u-1)(uv-1)}{(v-1)^2}C_A$	Literal equation in continuation: 1012.3419 Eq. (30), source label evlH	Continuation; same explicit map.	Same target qualifications.
C15 new expression: $E_N = -\frac{C_A(u-1)(uv-1)}{(v-1)^2} - 2C_F(u-1)$					
1012.3419 Eq. (30), source label evlH source $\frac{\partial \hat{H}(z_h, \mu^2)}{\partial \ln \mu^2} = \frac{\alpha_s}{2\pi} \int \frac{dz}{z} \left[A(\hat{z}) \hat{H}(z, \mu^2) + \int \frac{dz_1}{z_1^2} \text{PV} \left(\frac{1}{\frac{1}{z} - \frac{1}{z_1}} \right) B(z_h, z, z_1) \hat{H}_F(z, z_1, \mu^2) \right],$					
C15 cont.	New expression in continuation.	$C_{\text{bare}} = -P_0/\epsilon + (-B + 2 \ln u)P_0 - P_1$	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C15 new expression: $C_{HF} = -(B - 2 \log u)K_N - E_N$					
C15 cont.	New expression in continuation.	$C_{\text{ct}} = P_0/\epsilon, \quad C_{HF}^{(1)} = (-B + 2 \ln u)P_0 - P_1$	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C15 new expression: $Z_{HF} = K_N/\epsilon$					
C15 cont.	New expression in continuation.	Same accepted comparison.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C15 new expression: $\Delta_{\text{two cuts}} = 0$					
C15 cont.	New expression in continuation.	Same accepted comparison.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C15 new expression: $\partial_B C_{HF} + K_N = 0$					
C15 cont.	New expression in continuation.	Same accepted comparison.	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C15 new expression: $[C_{HF}]_{\epsilon^{-1}} = 0$					
C15 provenance. 7 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. All archived fields of this group are matched. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. The displayed color-collected/finite-relation form is independently checked against these unchanged native InputForms with exact zero difference; expanded forms remain linked. Historical group status: UNVERIFIED. Historical conversion: Fresh FeynArts currents, full Wilson reduction, independent physical traces/hat pieces, conjugate cuts, separate master integrations and actual collinear subtraction yield the displayed finite relation. The finite/RG/route identities are internal; K11 evolution and YZ tail do not publish this complete finite TMD HF coefficient. New-minus-old: 0. Complete literature difference is unverified. C15 source/export bindings ; executable comparison .					

Quantity / order	New generated expression	Accepted-engine expression	Exact reference equation / citation	Explicit conversion	Check result
C16 Connected coefficient ver- sus both literal alternatives	New expression in continuation.	$B_N = 2u(uv - u + 1)C_F + \frac{u(uv^2 + u - v - 1)}{(v - 1)(uv - 1)}C_A$	Literal equation in continuation: 0903.4680 Eq. (7)	Each native field preserves its original InputForm/LaTeX. Explicit function/dummy-name maps: {}. No paper-equation equality is inferred from an old/new field match.	Old: UNVERIFIED; paper: UNVERIFIED
C16 new expression: $\mathcal{B}_D = B_N + \epsilon E_B$					
0903.4680 Eq. (7) source $A = C_F \left[\left(z^3 \frac{\partial}{\partial z} \frac{\hat{H}(z)}{z^2} \right) (-2\hat{\xi}) + \hat{H}(z) \frac{2\hat{\xi}^2}{(1 - \hat{\xi})_+} \right] + \int \frac{dz_1}{z_1^2} PV \left(\frac{1}{\frac{1}{z} - \frac{1}{z_1}} \right) \hat{H}_F(z, z_1)$ $\times \left[-C_F \frac{2z_h}{z} \left(1 + \frac{z_h}{z_1} - \frac{z_h}{z} \right) - \frac{C_A}{2} \frac{2z_h}{z} \frac{zz_1(z + z_1) - z_h(z^2 + z_1^2)}{z(z - z_1)(z_1 - z_h)} \right].$					
C16 cont.	New expression in continuation.	$K_N = B_N/(1 - v)$	1012.3419 Eq. (25), source label bz source $B(z_h, z, z_1) = C_F \left[\frac{2z_h}{z} \left(1 + \frac{z_h}{z_1} - \frac{z_h}{z} \right) \right] + \frac{C_A}{2} \left[\frac{2z_h z_h^2(z^2 + z_1^2) - zz_1(z + z_1)}{z(z_1 - z)(z_1 - z_h)z} \right],$	Continuation; same explicit map.	Same target qualifications.
C16 new expression: $B_N = \frac{C_A u(uv^2 + u - v - 1)}{(v - 1)(uv - 1)} + 2C_F u(uv - u + 1)$					
C16 cont.	New expression in continuation.	$\Delta_K = -\frac{C_A u^2(v^2 + 1)(zh - 1)}{(v - 1)(uv - 1)}$	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C16 new expression: $E_B = \frac{C_A(u - 1)(uv - 1)}{v - 1} + 2C_F(u - 1)(v - 1)$					
C16 provenance. 1 fresh native field comparisons to actual archived native operator exports, executed in Wolfram after fresh production. Only a subset is matched; the whole group remains UNVERIFIED. Omitted scope: C16 normalization factors have different grouping: new factors omit mirror2 and sum actual two cuts separately. Shared Clifford/Wilson/spectral operator algorithms and external defining assumptions remain explicit. Final native binding is recorded separately. The displayed color-collected/finite-relation form is independently checked against these unchanged native InputForms with exact zero difference; expanded forms remain linked. Historical group status: MISMATCH. Historical conversion: Independently frozen recursion/Fierz/Wilson algebra equals fresh full-D graphs and the native bracket. Native versus converted YZ is exact; native versus literal K11 has the CA-only residual. Full measure is du/u dv/(1-v), so the density residual includes one additional (1-v)^-1. New-minus-old: UNVERIFIED; no machine-readable old target bound. Complete literature difference is unverified. C16 source/export bindings ; executable comparison .					
Fresh C16 bulk subchecks. The actual final mixing density gives $B_N = (1 - v)K_N$. In the separately stated $u = z_h/z$, $v = z/z_1$ map, native minus converted YZ is zero, while native minus the literal K11 bracket is					
$-\frac{C_A u^2(v^2 + 1)(zh - 1)}{(v - 1)(uv - 1)}.$					
This is a MISMATCH of the K11 bulk subexpression, with a zero C_F residual and the displayed nonzero C_A residual. The literal printed equation is unchanged in its table cell. For the admissible test $\Phi = v(1 - u)^2(1 - v)^2(-uv + 1)$, the exact full-density action is					
$\int_0^1 dv \Delta K \Phi = -\frac{3C_A u^2(u - 1)^2(zh - 1)}{4} \neq 0$					
at fixed $0 < u < 1$, $0 < z_h < 1$. This counterexample survives the physical double-zero endpoint restriction. Whole-group C16 remains UNVERIFIED where normalization grouping or auxiliary fields have not all been compared. Executable source/export and literal snippet bindings: <code>audit/c16-final-001/C16_LITERAL_SUBCHECKS.json</code> .					
C17 Generic end-point coefficient values beyond the physical class	UNVERIFIED: no new expression mapped.	c_0, c_1 undetermined on generic fields	1012.3419 Eq. (10) source $\hat{H}_D(z_1, z_2) = PV \left(\frac{1}{\frac{1}{z_1} - \frac{1}{z_2}} \right) \hat{H}_F(z_1, z_2) + \delta \left(\frac{1}{z_1} - \frac{1}{z_2} \right) \frac{z_1}{z_2} \hat{H}(z_2),$	No new expression has been mapped to this equation group. Its accepted expression and historical map are retained for inspection.	Old: UNVERIFIED; paper: UNVERIFIED
C17 cont.	Same new-route scope.	$(c_0, c_1) = (0, 0)$ is only a physical-class representative	No additional published counterpart.	Continuation; same explicit map.	Same target qualifications.
C17 provenance. Not established by the current new-route comparison record. Historical group status: UNVERIFIED. Historical conversion: The fresh final transport contains Missing generic coefficients and explicitly labels the zero representative. A pointwise kernel or quotient identity does not determine generic c0/c1. C17 source/export bindings ; executable comparison .					

Pinned references and source identities

Each identifier below is the source version used by the retained literal ledger. Source hashes identify archive bytes; the ledger separately records equation blocks and macro hashes. The pp hadron-in-jet hard factors are not ep hard references. **EP: 2007.07281v1** <https://arxiv.org/abs/2007.07281v1>

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